Predicting Chronic Disease Onset in Older Adults: A Machine Learning Study using The Irish Longitudinal Study on Ageing (TILDA)

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Abstract

Background: Chronic disease and functional decline are major health challenges for older adults, yet how well self-reported health burden can predict disease onset across longitudinal waves remains underexplored.

Methods: Using five waves of data (2009–2019), a longitudinal machine learning framework evaluated two tasks: predicting any chronic disease onset across three designs (single wave, two pooled multi-wave), and 24 disease specific outcomes. Four model families were compared via cross-validation (CV) area under the curve (AUC), with predictor contributions assessed by bootstrapped add-on analysis.

Results: Pooled multi-wave modelling improved by 22% over the single wave baseline (0.718 ± 0.006 vs 0.589 ± 0.072), with cross-fold variance decreasing more than tenfold. Disease specific models achieved test performance up to 0.721 for functional outcomes. Reducing to the 20 most important features caused no measurable performance loss, while cognitive, psychosocial, nutritional, and behavioural add-on predictors showed no statistically significant improvement.

Conclusions: Among community dwelling older adults, chronic disease risk prediction was primarily driven by self-reported burden indicators, particularly medication use, pain, and self-rated health rather than domain specific biomarker or lifestyle profiles. These findings suggest that publicly available survey data can support meaningful risk stratification with a limited feature set.

Keywords: ageing, Machine Learning (ML), predictive modelling, chronic disease, functional decline, health burden, The Irish Longitudinal Study on Ageing (TILDA), Extreme Gradient Boosting (XGB).

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AI Declaration

I acknowledge the use of GitHub Copilot Student to assist with code completion, debugging, and LaTeX formatting during the preparation of this thesis. I also acknowledge the use of Gemini, Microsoft Copilot, Grammarly, and ChatGPT to support general brainstorming, writing preparation, literature organisation, and LaTeX formatting support.

Prompt examples are included in the auxiliary supporting material submitted alongside the final submission of this study.

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List of Abbreviations

ADL	Activities of Daily Living
AI	Artificial Intelligence
AP	Average Precision
AUC	Area Under the ROC Curve
BMI	Body Mass Index
BP	Blood Pressure
brainPAD	Brain predicted age difference
CAPI	Computer Assisted Personal Interview
CES-D	Center for Epidemiologic Studies Depression Scale
CI	Confidence Interval
CRP	C-Reactive Protein
CSV	Comma Separated Values
CV	Cross Validation
DM	Diabetes Mellitus
DNN	Deep Neural Network
EU	European Union
FI	Frailty Index
FOF	Fear of Falling
FR	Frailty
G2G	Gateway to Global Ageing Data
GBM	Gradient Boosting Machine
GDPR	General Data Protection Regulation
GRIP	Grip Strength
GSR	Gait Speed Reserve

List of Abbreviations

HADS	Hospital Anxiety and Depression Scale
HGS	Hand Grip Strength
HPO	Hyperparameter Optimization
HRS	Health and Retirement Study, (a longitudinal study of ageing in the United States)
IADL	Instrumental Activities of Daily Living
ISSDA	Irish Social Science Data Archive
JSON	JavaScript Object Notation
LR	Logistic Regression
MGS	Maximum Gait Speed
MH	Scales and Questions on Mental Health and Wellbeing
ML	Machine Learning
MLP	Multi-Layer Perceptron
MMSE	Mini-Mental State Examination
MOCA	Montreal Cognitive Assessment
MRI	Magnetic Resonance Imaging
MVPA	Moderate-to-Vigorous Physical Activity
NOS	Newcastle-Ottawa Scale
OR	Odds Ratio
PA	Physical Activity
PC1	First Principal Component
PCA	Principal Component Analysis
PMF	Pseudonymised Microdata File
PR-AUC	Precision Recall Area Under the Curve
PRISMA	Preferred Reporting Items for Systematic reviews and Meta Analyses
RAND	Research and Development Corporation
RF	Random Forest
ROC	Receiver Operating Characteristic
SCQ	Self Completion Questionnaire
SHAP	Shapley Additive Explanations
SHARE	Survey of Health, Aging and Retirement in Europe

SOC	Social Variables
SPSS	Statistical Package for the Social Sciences
TILDA	The Irish Longitudinal Study on Ageing
TUG	Timed Up and Go (mobility test)
UCLA	University of California Los Angeles
UGS	Usual Gait Speed
UL	University of Limerick
WHO	World Health Organization
XGB	Extreme Gradient Boosting

List of Abbreviations

Chapter 1

Introduction

Ageing populations experience substantial changes across physical, nutritional, cognitive, and psychosocial domains, each shaping vulnerability to chronic disease and functional decline [6–8]. Understanding how these multidomain factors interact over time is essential for early risk identification.

Lifestyle risk factors are estimated to contribute over $514 billion in direct and indirect healthcare costs annually across Europe [9], and in Ireland, malnutrition alone is estimated to cost over €1.4 billion per year in hospital and community care costs [10]. Frameworks such as the World Health Organization (WHO) *Global Action Plan* and Ireland's *Healthy Ireland* strategy identify modifiable health behaviours as central to disease prevention. European regulatory frameworks, including the General Data Protection Regulation (GDPR) and the proposed European Union (EU) Artificial Intelligence (AI) Act, further emphasise the need for transparent and interpretable Machine Learning (ML) models in healthcare [11].

Traditional risk prediction models have largely relied on clinical biomarkers such as Blood Pressure (BP) and lipid profiles. Yet, growing evidence shows that broader predictors, including nutrition [12], Physical Activity (PA) [13], Scales and Questions on Mental Health and Wellbeing (MH) [14], and Social Variables (SOC) [15] each demonstrate independent associations, improving predictive scope.

The Irish Longitudinal Study on Ageing (TILDA)[1] is a nationally representative longitudinal study of community dwelling Irish adults aged 50 or older, with data collected repeatedly across physical, clinical, functional, psychological, and social domains. An externally produced Research and Development Corporation (RAND) harmonised TILDA dataset (waves 1–2, the only waves currently harmonised), aligned to international standards, was inspected and preprocessed alongside the five core waves.

Despite these advantages, existing ML research on chronic disease prediction faces persistent limitations. Models over rely on clinical predictors, rarely incorporate lifestyle and SOC, predominantly use cross sectional designs that fail to capture risk trajectories, and give insufficient attention to the preprocessing and representation of heterogeneous longitudinal health data.

This study addresses these gaps by applying ML to pooled and cross-wave TILDA data, evaluating whether richer predictor sets, longitudinal designs, and reproducible preprocessing improve chronic disease onset prediction in older adults.

1.1 Research problem and motivation

Despite increasing research and clinical interest in lifestyle, nutritional factors, and advances in ML methods, approaches to chronic disease prediction in older adults remain limited in several ways.

- **Limited integration of multidomain predictors:** Existing models focus primarily on clinical measurements and often neglect modifiable behavioural, dietary, and psychosocial factors.

- **Underuse of longitudinal dynamics:** Few studies use repeated measures to track how risk changes over time, such as the progression of Frailty (FR) [6]. Probabilistic reasoning can help estimate the likelihood of developing a condition at a future time given earlier measurements, for example, calculating the probability $P(\text{FR at wave 5} \mid \text{UGS at wave 1}, \text{Nutrition at wave 2})$.

[1] TILDA's data are pseudonymised and accessed under license via Irish Social Science Data Archive (ISSDA). Use adheres to University of Limerick (UL) Science & Engineering ethics approval and data retention requirements.

- **Healthcare data preprocessing:** Longitudinal health survey data present significant representation challenges. Variables encoding the same concept may differ in format, value ordering, and coding scheme across waves, while health specific constructs such as frailty index items require careful ordinal handling to preserve clinical meaning during numerical conversion [16]. These challenges are rarely reported in the ML literature, yet the quality of downstream predictions depends directly on how accurately the data are prepared.

These gaps motivate the multi-wave, multidomain approach taken in this study, which uses TILDA's five longitudinal waves to test whether richer predictor sets and pooled designs improve chronic disease onset prediction.

1.2 Research aims and objectives

The research develops and evaluates longitudinal ML approaches that combine multidomain predictors from TILDA to predict chronic disease onset in older adults, with a focus on identifying the features and modelling designs that best support this task. This goal will be achieved through the following objectives:

- Review existing applications in ML with TILDA for chronic disease prediction in ageing populations, with emphasis on gaps in lifestyle and nutrition integration.

- Develop a reproducible pipeline to preprocess and align TILDA waves 1–5, deriving multidomain features across physical, functional, clinical, lifestyle, and psychosocial domains.

- Develop and compare a linear baseline (Logistic Regression (LR)) against tree ensemble and neural network families (Extreme Gradient Boosting (XGB), Random Forest (RF), Multi-Layer Perceptron (MLP)) across single wave and pooled longitudinal designs.

- Evaluate models using Area Under the ROC Curve (AUC), calibration, and classification metrics across all designs.

- Identify the most informative predictors of chronic disease onset and, through a predictor add-on analysis, assess whether domain specific features contribute independent predictive value beyond the core health burden set.

- Document ethical, governance, data access, and reproducibility practices for the pseudonymised longitudinal data used in this study.

1.3 Contributions of the research

This research makes contributions at three levels:

- **Empirical:** Demonstrates that longitudinal pooling substantially improves predictive performance and stability for chronic disease onset prediction in this cohort, and identifies the compact set of health burden indicators that drive this signal. Nutritional, psychosocial, and behavioural domain features added no independent predictive value beyond this core set, a finding discussed in Chapter 5.

- **Methodological:** Provides a reproducible pipeline for aligning longitudinal TILDA data across domains and integrating features into ML workflows. Full pipeline implementation, including the cleaning engine, scenario detection scripts, and modelling notebooks, is available at `https://gitlab.com/abderahman-msc-project/milestones/-/blob/main/README.md` (private repository, access granted on request).

- **Clinical:** Identifies the most informative predictors of chronic disease onset and evaluates the marginal contribution of domain specific features, providing a data driven basis for risk stratification. These findings could support healthcare providers in identifying high risk individuals earlier and informing targeted prevention strategies, and may help older adults understand which observable health indicators are most relevant to them, respectively.

1.4 Thesis outline

Chapter 2 reviews ML approaches to chronic disease prediction in ageing populations, with emphasis on lifestyle, nutrition, and methodological considerations. *Chapter* 3 describes the data preparation process, including the TILDA study design, data collection, thematic domain structure, and the manual variable screening workflow. In *Chapter* 4, the automated cleaning pipeline is presented, including the variable scenario detection system, the deterministic six step cleaning sequence, and validation of the pipeline against published benchmarks. *Chapter 5* presents the empirical evaluation, covering performance across three modelling designs, disease specific prediction, predictor add-on analysis, Shapley Additive Explanations (SHAP) based interpretability analysis, and an interpretation of the health burden correlation structure underlying the results. *Chapter 6* discusses the implications of the findings and future directions for the research.

Chapter 2

Literature Review

This chapter reviews literature on ageing, chronic disease, and predictive modelling using TILDA covering four interrelated domains: physical function and FR, MH, nutrition and biological ageing, and ML applications for chronic disease prediction.

2.1 Search strategy and study selection

This review applied systematic mapping principles [1]. Research questions were defined, relevant databases searched, inclusion and exclusion criteria applied, and findings synthesised through an integrative approach [2, 17]. The primary focus was chronic disease prediction in older adults using TILDA. Where ML applications specific to TILDA were limited, the scope was extended to include comparable ageing cohort studies.

2.1.1 Database search and query development

The main topic of interest was chronic disease onset prediction in older adults using ML, with TILDA as the primary data source. Relevant databases (e.g., ScienceDirect and IEEE Xplore) were searched using combinations of keywords (i.e., "Irish Longitudinal Study on Ageing", "machine learning", "chronic disease", "nutrition", "diet", and "physical activity"), as shown in Table 2.1.

The search covered studies published from 2011 to 2025. Figure 2.1 summarises the number of studies identified per year, highlighting trends and gaps over time.

Table 2.1: Database searches and number of studies per database [1].

Database	Search	Results
Oxford Academic	(TILDA OR Irish Longitudinal Study on Ageing) AND (chronic disease OR diabetes OR cardiovascular) AND (nutrition OR diet OR "physical activity")	1081
	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("machine learning" OR "predictive model" OR "risk prediction")	71
IEEE Xplore	TILDA OR "Irish Longitudinal Study on Ageing"	56
	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("machine learning" OR "predictive model" OR "risk prediction")	7
ScienceDirect	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("chronic disease" OR cardiovascular OR diabetes) AND (nutrition OR diet OR lifestyle OR "physical activity")	257
	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("machine learning" OR "predictive model") AND ("chronic disease" OR diabetes) AND (nutrition OR "physical activity")	20
PubMed	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("chronic disease" OR cardiovascular OR diabetes) AND (nutrition OR diet OR lifestyle OR "physical activity")	27
	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("machine learning" OR "predictive model" OR "risk prediction")	9

Some search strings returned few or irrelevant entries, particularly for ML applications in TILDA, highlighting a key research gap. Iterations continued until suitable combinations consistently retrieved relevant studies. Duplicates were removed and the remaining studies screened individually for relevance to ageing, chronic disease, and ML. Key elements for each study, including model type, features used, and main findings, were recorded in Table 2.2. The full selection process is illustrated in the Preferred Reporting Items for Systematic reviews and Meta Analyses (PRISMA) diagram (Figure 2.2).

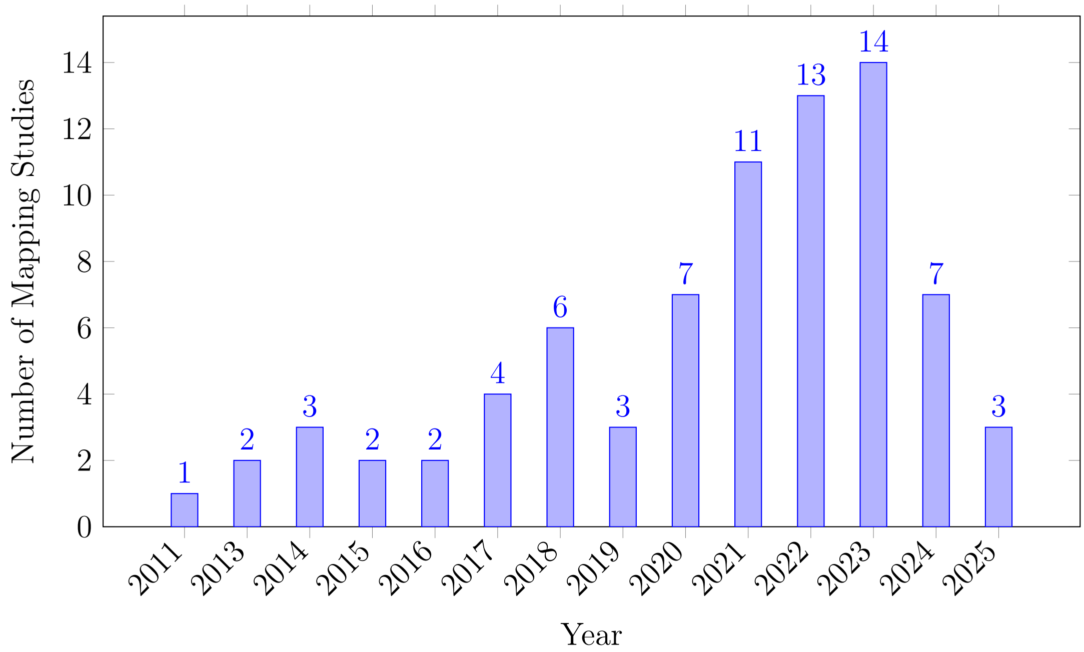

Figure 2.1: Publications per year.

2.1.2 Quality and relevance evaluation

A customised version of the Newcastle-Ottawa Scale (NOS) was applied to evaluate methodological qualities across the included studies, illustrated in Table 2.3. The NOS evaluates studies across three domains: *selection* (S1–S4), *comparability* (C1–C2), and *outcome* (O1–O3). Selection criteria assess the representativeness of the sample, clarity of exposure measurement, and temporal alignment of outcomes. Comparability examines the control of key confounders (variables that may influence both the predictor and outcome), while outcome evaluates measurement validity, follow up adequacy, and completeness. Each domain is scored binary (0 = criterion not met, 1 = criterion met), and the total reflects overall study quality.

This initial table guided refinement into two subsets. Studies critically related to TILDA, and studies applying ML within TILDA for predictive modelling. The

Table 2.2: Summary of selected studies on ageing using TILDA.

Theme	Author/year	Dataset	Variables/features	Methods/model type	Key Findings/outcome	Limitations
Frailty (FR)	Romero-Ortuno (2021) [18]	Wave 1–5	FR state, FR phenotype (FP) components (Exhaustion (EX), Unexplained weight loss (WL), Weakness (WK), Slowness (SL), Low physical activity (LA))	Multi state Markov model, longitudinal 8 year follow up	32% of pre frail to non frail; FR strongly predicted 31% mortality	Self report bias; observational design
Falls + Machine Learning (ML)	Donoghue (2023) [19]	Wave 1–3	Gait, mobility, medications, history of falls	Conditional inference forest; 5-fold Cross Validation (CV); undersampling for imbalance	Accuracy up to 69.7%; for adults aged 50–64.	AUC values were ≤ 0.69 and Precision Recall Area Under the Curve (PR-AUC) ≤ 0.39 indicating low predictive accuracy; no external validation; cross sectional; Class imbalance
Fear of Falling (FOF)	Zarudsky (2014) [20]	Wave 1	Age, sex, fall history	Cross sectional prevalence analysis	23.3% prevalence; higher in women; 70% with Fear of Falling (FOF) had no recent falls in the last 12 months	Self-reported; cross sectional
Functional Mobility	Huang (2019) [21]	Wave 1–2	Measured by Timed Up and Go (mobility test) (TUG), Hand Grip Strength (HGS), physical activity, vision, Body Mass Index (BMI), age	Linear regression, stratified analyses	Age modifies effects; PA and grip protect mobility; vision benefit only in middle aged (Beta = -0.032, P = 0.002)	Observational; 2 year follow up
Depression	Jacob (2023) [22]	Wave 1–2	Falls, FOF, HADS-A, Center for Epidemiologic Studies Depression Scale (CES-D)	Prospective LR	Falls predicted incident anxiety Odds Ratio (OR)=1.58 and depression OR=1.43	Self-reported falls; short follow up; mediation effect partially explored
	Laird (2023) [23]	Wave 1–5	Moderate-to-Vigorous Physical Activity (MVPA), walking, sociodemographics	Longitudinal regression; mixed model	Higher physical activity reduced depressive symptoms; dose response effect	Self-reported PA; observational; limited causal inference
Nutrition	Murphy (2023) [12]	Wave 1–5	Plasma lutein, zeaxanthin	Linear/ordinal LR	Higher carotenoids linked to slower FR progression Lower odds of becoming frail (ORs = 0.57–0.89)	Observational; not predictive of longitudinal muscle changes
	Bardon (2020) [24]	Wave 1–2	Hospitalisation, functional status, SOC support, sex, falls	LR; sex stratified analyses	Functional decline, hospitalisation, poor support increased malnutrition risk; sex differences	Observational (short follow up (2 years)); limited generalisability >65 years, self-reported
ML	Xu (2023) [25]	Wave 1–3	105 predictors from 40,000+ variables (SOC, clinical, mental, family)	Stacked ensemble: RF, Extreme randomised trees (XRT), General linear model (GLM), Gradient Boosting Machine (GBM), Deep Neural Network (DNN); SHAP explainability	AUC=0.854 for 2 year Diabetes Mellitus (DM) prediction; top predictors LDL, cirrhosis, sex	No external validation; short follow up; ensemble complexity limits deployment
	Davis (2021) [7]	Wave 3	Age, HGS, chair stands, BMI, cognitive tests	Histogram gradient boosting regression; SHAP	r^2 test: UGS≈0.41, Maximum Gait Speed (MGS)≈0.46, GSR≈0.21	Modest predictive power; cross sectional; no external validation
	Zhang (2025) [26]	Longitudinal IMAGEN cohort	Psychopathology, personality, cognition, substance use, environment	Regularised LR; diagnostic	Longitudinal AUC: ED=0.71, MDD=0.64, AUD=0.67; shared transdiagnostic predictors	Limited age range; no external replication
	Boyle (2021) [27]	Wave 3	Brain predicted age difference (brainPAD); processing speed; attention; cognitive flexibility; verbal fluency	Linear regression, correlation analyses	Higher brainPAD associated with lower cognitive performance across multiple domains	Cross sectional; no causal inference; limited sample

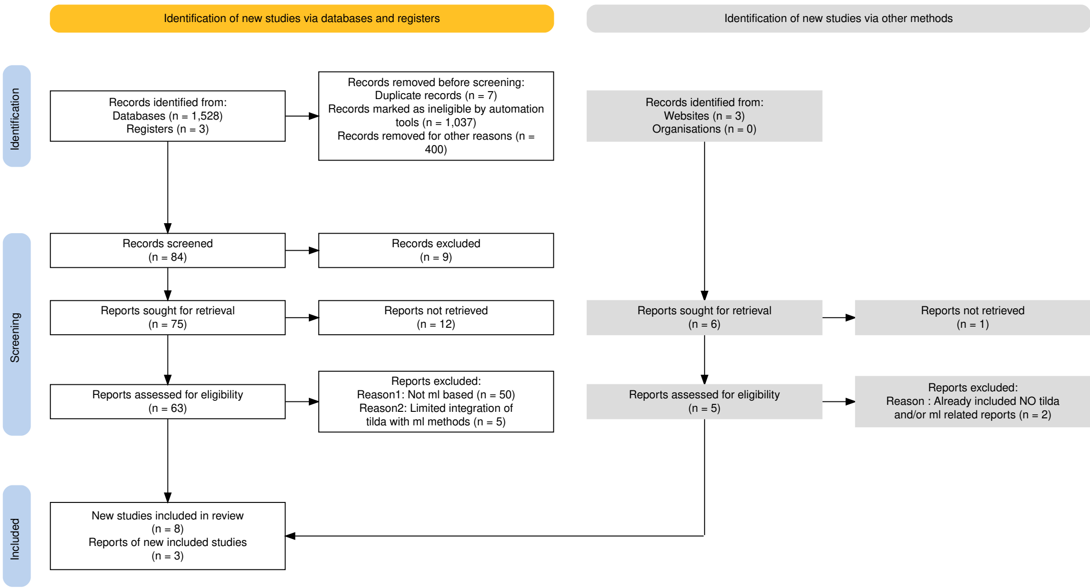

Figure 2.2: PRISMA chart for systematic reviews showing the study selection process for the systematic mapping and integrative review [5].

second subset revealed minimal integration between ML methods and TILDA. Data were extracted and synthesised to identify methodological trends, including Markov models, regression baselines, and CV protocols [2]. Given the limited number of ML studies using TILDA directly, the review was broadened to include chronic disease prediction research in comparable older adult populations.

2.2 Physical function and frailty

FR reflects declining physiological reserve, providing a framework for understanding physical vulnerability in ageing. In TILDA, it is captured through both the FR phenotype and the FR index, with UGS, HGS, and mobility decline serving as practical predictors. These measures strongly predict hospitalisation and mortality [18].

At the same time, FR indicators are highly correlated with one another. When Grip Strength (GRIP) declines, gait speed and functional capacity typically follow, meaning each additional predictor captures diminishing independent information. This correlation structure has direct implications for ML models, which

Table 2.3: Quality evaluation of included systematic literature reviews [2].

Reference	S1	S2	S3	S4	C1	C2	O1	O2	O3	Total
(Xu, 2023 [25])	1	1	1	1	1	1	1	1	1	9
(Donoghue, 2023 [19])	1	1	1	1	1	1	1	0	1	8
(Moguilner, 2021 [6])	1	1	1	1	1	0	1	1	1	8
(Davis, 2021 [7])	1	1	1	1	1	0	1	1	0	7
(Boyle, 2021 [27])	1	1	1	1	1	0	1	0	1	7
(Tzemah-Shahar, 2022 [28])	1	1	1	0	1	0	1	1	1	7
(A. Shirsath, 2024 [29])	1	1	1	1	1	1	1	0	1	8
(Romero-Ortuno, 2021 [18])	1	1	1	1	1	1	1	1	1	9
(Bardon, 2020 [24])	1	1	1	0	1	0	1	1	1	7
(Zhang, 2025 [26])	1	1	1	1	1	1	1	0	1	8
(Sutin, 2023 [30])	1	1	1	1	0	0	1	1	1	7

must decide how many such indicators to include before results become insignificant, a question revisited empirically in Chapter 5.

2.2.1 Probabilistic interpretation of frailty

From a probabilistic perspective, the likelihood of an adverse outcome Y (such as mortality) given observed FR features X is expressed as a conditional probability $P(Y \mid X)$, and identifying which features carry the strongest predictive signal requires understanding how much unique information each contributes.

When features X_i and X_j are statistically independent,

$$P(X_i, X_j) = P(X_i)\, P(X_j),$$

and each adds fully independent information to the model. In practice, however, FR related predictors such as UGS, HGS, and mobility limitation are often highly correlated, so the features share information and the additional gain from including a second correlated predictor is reduced.

Evidence suggests that FR is dynamic, not irreversible. For example, multi state modelling showed that many prefrail adults transition back to non frail, although established FR strongly predicts mortality [18]. Metabolic syndrome accelerates the development of FR, suggesting a metabolic pathway of vulnerability [31]. The presence of metabolic syndrome raises $P(\text{FR} \mid \text{Metabolic Syndrome})$

and is associated with greater risk of adverse outcomes over time [31]. Age also plays an important role. Obesity predicts mobility decline earlier in life, whereas later in life, traits that protect against this, including HGS, become more significant [21]. The rate of BP recovery after postural change has also been linked to accelerated brain ageing in TILDA [29], illustrating that physiological vulnerability in this cohort extends beyond conventional FR markers into cardiovascular dynamics.

These findings partially conflict when FR appears both reversible and progressive, depending on risk context. This complexity motivates probabilistic models that can update beliefs about risk as new observations arrive, as formalised by Bayes theorem:

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}$$

where $P(Y \mid X)$ is the posterior probability of outcome Y given features X. In practice, causal inference remains limited by observational designs. This probabilistic framing motivates the use of well calibrated ML models evaluated via AUC, calibration metrics such as the Brier score, and explainability tools such as SHAP.

2.3 Mental health in older adults

MH, including depression, anxiety, loneliness, and SOC isolation, strongly shapes quality of life. Evidence from TILDA shows strong links to both physical and SOC factors. Participation in physical, SOC, and cognitive activities, summarised by the Act-Belong-Commit (ABC) indicator, was associated with lower depression and anxiety, suggesting that SOC integration and lifestyle protect mental wellbeing [32]. However, self-reported measures may overestimate such associations, raising concerns about response bias. These two findings show some disagreement. The effect of SOC engagement is strong when self-reported and weaker when objectively assessed.

Objective data strengthen the case for activity. Accelerometer based measures show a dose response link between MVPA and lower depressive symptoms [13]. Yet reverse pathways are also evident, incident falls and functional decline predict later depression, mediated by FOF [22]. Together, these results reveal a bidirectional relationship, MH both influences and is influenced by physical decline. Most studies rely on broad self report scales, such as CES-D or Hospital Anxiety and Depression Scale (HADS), with little biomarker or clinical validation.

This literature suggests MH in ageing requires longitudinal, multidimensional approaches capable of capturing bidirectional pathways. Formally, the conditional probability

$$P(\text{Depression} \mid \text{Activity, Falls, SOC Engagement})$$

quantifies how these interacting features jointly influence MH outcomes, indicating which predictors carry the strongest signal and which may be redundant.

2.4 Nutrition and biological ageing

Nutrition interacts with both biological and psychosocial systems, shaping ageing trajectories and FR. Biomarkers such as vitamin D, lutein, and zeaxanthin have been linked to physical and cognitive outcomes in TILDA. Higher plasma lutein and zeaxanthin, dietary carotenoids found in leafy vegetables, were associated with slower FR progression [12], while vitamin D deficiency predicted DM and musculoskeletal decline, particularly among adults with low sunlight exposure [33]. These studies point to two distinct nutritional mechanisms, antioxidant protection and metabolic regulation. Whether plasma biomarkers improve prediction beyond self-reported health indicators is tested directly in Chapter 5.

2.4.1 Social context and nutritional risk

Malnutrition also reflects SOC and functional risks. Hospitalisation, functional loss, and poor SOC support were found to increase malnutrition risk, with notable sex differences [24], highlighting that biological markers are often studied without integrating SOC context.

The difference between biological and chronological age remains a major point of debate. Physical capacity measures such as UGS and GRIP have also been proposed as accessible proxies for biological ageing [28]. Biomarker based metrics that better capture these cumulative exposures, like FR indices or epigenetic clocks, frequently vary from chronological age. McCarthy links metabolic syndrome and nutritional deficits to accelerated biological ageing, as indexed by epigenetic clocks [34]. Although most of this evidence is cross sectional and cannot capture change over time, these indicators offer a more precise measure of vulnerability. Reliability is also reduced in biomarker studies with small sample sizes, limiting confidence in the estimated effect magnitudes.

Nutrition interacts with biological, metabolic, and SOC systems simultaneously. Studying one nutrient at a time risks missing this bigger picture. Integrative modelling that combines biomarkers, clinical factors, and SOC determinants supported by ML pipelines offers a more promising route. Models that estimate the joint distribution of features and outcomes can better capture the complex dependencies relevant to biological ageing.

2.5 Machine learning for chronic disease prediction

ML is increasingly applied in ageing research because it can capture nonlinear and multidimensional relationships that traditional regression models, such as LR, often miss [25, 27]. Studies using ML in ageing research include many types of predictors: clinical measures, sociodemographic factors, lifestyle habits, cognitive tests, and psychosocial variables.

Models consistently favour ready measured clinical data such as comorbidity counts, mobility scores, and vital signs, while nutrition, SOC, and lifestyle predictors remain underrepresented [6, 25]. This imbalance may limit the ability of models to capture the full complexity of chronic disease risk with ageing, especially given that MH and nutrition exert independent and interacting effects on later life outcomes.

2.5.1 Methodological approaches

The methodological approaches used in ageing research span both traditional and advanced models. LR and Markov models remain common baselines due to their interpretability and simplicity, yet they are often unable to capture nonlinearities or higher order interactions. More recent work has shifted towards ensemble methods such as RF, GBM, as well as DNNs, which generally achieve higher predictive accuracy. For instance, with TILDA, LR based approaches have achieved an AUC of 0.78 for mortality prediction [35], while a stacked ensemble combining gradient boosting and neural networks reached 0.854 for two year DM onset prediction [25].

2.5.2 Performance and validation

Predictive performance across studies spans a wide range, with AUC values between 0.64 and 0.85 for outcomes including MH, diabetes, falls, and mortality. This spread raises a question worth asking. Why does performance vary so much between what appear to be similar tasks? One underappreciated reason is how health data are preprocessed before modelling. Longitudinal survey variables often contain mixed formats, inconsistent coding, and wave specific encodings that, if handled incorrectly, can silently distort model inputs. Studies rarely report preprocessing decisions in enough detail to assess whether performance differences reflect genuine signal or encoding artefacts. This reproducibility gap is a methodological limitation that this study directly addresses through the deterministic cleaning pipeline described in Chapter 4.

Most studies are cross sectional, limiting any causal or temporal assessment. Narrow predictor sets and observational designs together constrain the generalisability of existing models [25].

2.5.3 Explainability in machine learning for ageing

As ML models become more complex, interpretability emerges as a critical challenge. Explainability tools like SHAP can highlight insights into the decision making process of a model, but their use is inconsistent across studies, particularly in deep learning models [4, 7, 25]. These methods improve transparency by revealing which features drive predictions, yet black box concerns persist, particularly in deep learning, where feature interactions are difficult to track. For ageing research, where clinical adoption requires both accuracy and trust, explainability is not optional but essential. Current efforts show promise but remain fragmented, with no consensus on best practice.

2.6 Gaps in the literature

Five gaps emerge from the reviewed literature that this study directly addresses. First, existing studies applying ML to TILDA have largely used single waves or short follow-up windows, limiting the ability to capture how risk changes across time [6, 7, 19, 25, 27]. This study uses all five available waves, enabling pooled longitudinal designs that test whether temporal structure improves prediction. Cross study replication using the RAND harmonised TILDA dataset, which maps variables to international standards compatible with cohorts such as Health and Retirement Study, (a longitudinal study of ageing in the United States) (HRS) and Survey of Health, Aging and Retirement in Europe (SHARE), is identified as a direction for future work.

Second, predictive models in ageing research overwhelmingly rely on physical health indicators such as comorbidity counts and mobility measures, while nutrition, SOC, and behavioural variables are rarely integrated. Nutritional and social factors independently influence health trajectories [12, 24]. Yet it remains an open empirical question whether these domains add predictive value beyond established clinical indicators in a general community sample. This study assesses that question directly through a predictor add-on analysis described in Section 5.3.4.

Third, the preprocessing and numerical representation of longitudinal health survey data is a genuinely difficult problem that receives little attention in the ML literature. Variables encoding the same concept may differ in format, value ordering, and coding scheme across waves. Health specific constructs such as Frailty Index (FI) items require ordinal aware handling to preserve clinical meaning, and incorrect encoding can silently distort downstream predictions. Few published studies describe their preprocessing workflow in sufficient detail to enable reproducibility or cross study comparison. This challenge is addressed through a transparent deterministic six step cleaning pipeline applied consistently across all TILDA waves, with fully traceable audit documentation.

Fourth, many outcomes relevant to older adults, including chronic pain and subjective wellbeing, remain underexplored as prediction targets. Standard focus on clinical diagnoses omits outcomes that are often more meaningful for intervention planning and quality of life. Psychological constructs such as sense of meaning and purpose have also been linked to incident dementia [30], yet rarely feature as prediction targets in ML ageing models. The research includes pain, functional decline, and hearing loss alongside incident cases (clinical) coded disease outcomes highlighted in Section 5.1.1.

Finally, where longitudinal designs have been used, cross wave validation is almost always internal. Weak validation and narrow feature sets limit the generalisability of existing models and increase the risk of overfitting. This study applies participant level train/test splitting with stratified group CV to prevent data leakage across all designs.

2.7 Chapter summary

The evidence reviewed in this chapter shows that FR, mental health, and nutrition do not operate in isolation. FR is partly reversible but remains strongly associated with mortality. MH both influences and is shaped by physical decline and nutrition interacts with biological, metabolic, and social systems in ways that single variable studies frequently underestimate.

While ML has improved prediction accuracy in ageing research, challenges remain in integrating diverse predictors and achieving robust longitudinal validation. An often overlooked challenge is the preprocessing of health survey data itself, where inconsistent formats and wave specific encodings can compromise model inputs if not handled carefully. These gaps motivated the integrative, longitudinal approach developed in subsequent chapters.

Chapter 3

Methodology

3.1 Study population and design

TILDA provides longitudinal data across waves for community dwelling adults in Ireland. The RAND harmonised TILDA dataset (waves 1–2), aligned to cross study standards via the Gateway to Global Ageing Data (G2G) framework, was also cleaned using the same pipeline.

All participants who provided data in at least one TILDA wave were considered eligible. To maximise the available sample for ML, no exclusions were applied on the basis of age, medical conditions, mobility, cognitive status, or health centre attendance.

Participants were removed only in the rare case that, following manual variable filtering, no analytically usable variables remained for that individual. Filtering occurred primarily at the *variable level* during the manual screening stage, excluding variables with extreme missingness, lack of variation, or non informative structure. This approach preserves the largest possible sample while ensuring all retained inputs are analytically meaningful.

3.2 TILDA data collection

TILDA data collection occurred over five longitudinal waves (see Table 3.1). Each wave included three primary components:

1. **Computer Assisted Personal Interview (CAPI):** Computer assisted personal interviews conducted in participants homes, covering health status, sociodemographics, economic circumstances, and family networks. This assisted format ensures automatic routing between questions, eliminates data entry errors, and allows real time consistency checks, improving data integrity compared with paper based methods.

2. **Self Completion Questionnaire (SCQ):** Paper based questionnaires completed privately by participants, capturing sensitive information including mental health, alcohol use, sexual activity, and subjective wellbeing. The self completion format reduces social desirability bias for topics participants may be reluctant to discuss with an interviewer.

3. **Health Assessment:** Objective clinical and biomarker measurements, including blood pressure, grip strength, gait speed, cognitive tests, and blood samples, conducted in dedicated health assessment centres (waves 1 and 3) or adapted for home measurement in subsequent waves.

All TILDA data were pseudonymised prior to release, meaning that personal identifiers such as names and addresses were removed or replaced with unique codes to protect participant confidentiality. Each participant received a unique identifier ('id'), along with household and cluster identifiers, allowing longitudinal linkage across waves without revealing individual identities.

Table 3.1: Dimensions of raw TILDA and harmonised datasets (version 5.5).

Dataset	Collection Period	Participants (N)	Variables (N)
Wave 1 [36]	2009–2011	8501	1619
Wave 2 [37]	2012–2013	7206	724
Wave 3 [38]	2014–2015	6397	1029
Wave 4 [39]	2016	5713	990
Wave 5 [40]	2018–2019	4978	983
Harmonised [41]	2016 (release)	8501	572

3.3 Thematic domains

Variables were organised into seven thematic domains based on ageing literature and study objectives. Each variable's prefix, as defined in the Pseudonymised Microdata File (PMF) metadata (Table 3.2), indicating its source or research domain. This systematic assignment ensures that each thematic group contains conceptually related features and provides a clear structure for downstream automated preprocessing and ML.

Home based CAPI variables use a two letter module code followed by a three digit question code. Variables with multiple responses or loops include an underscore and number to specify the response or iteration.

Additional derived variables, constructed from CAPI or SCQ responses, are documented in a separate variable codebook that specifies their source items and scoring logic. When all CAPI question branches and response iterations are fully expanded, wave 1 alone includes over $40,000$ parameters [25], of which the $1,619$ structured columns in Table 3.1 represent the column space available in the public release. These derived variables were assigned thematic domains based on the research area of their component items.

Categorising the raw variables into seven thematic domains provides a structured foundation for the dimensionality reduction and cleaning pipeline described in Section 3.4 and Chapter 4.

3.4 Manual variable screening

Prior to any data cleaning or modelling, a structured multi-stage workflow was implemented to screen and reduce the dimensionality of the raw TILDA datasets. As shown in Table 3.3 (for example, 1619 $\rightarrow$ 809 variables in wave 1), the raw PMF files were reduced to a subset of variables of interest. This procedure also separated automated diagnostic profiling from manual variable exclusion, ensuring transparency in all selection decisions. The process consisted of the following steps:

1. **Raw data loading:** Raw TILDA PMF files were loaded without cleaning or recoding. Due to compatibility issues, the harmonised dataset was loaded

Table 3.2: PMF v5.5 derived variable prefixes [3].

Variable Prefix	Research Area
1: Socio Demographic & Context	
INC	Income variables
SES	Social class
SOC	Social
2: Physical Health & Clinical Baseline	
BP	Blood pressure from sphygmomanometer
MD	Medication data
SCR	Screening tests
3: Functional Status & Frailty	
ADL	Activities of Daily Living (ADL)
DIS	Disability, functional impairment, helpers
FR	Frailty, gait, exercise, falls and fractures
GRIP	Grip strength
IADL	Instrumental Activities of Daily Living (IADL)
4: Lifestyle & Behavioural	
BEH	Health behaviours
5: Nutrition & Metabolic Biomarkers	
BLOODS	Results of blood sample analysis
6: Psychological & Cognitive	
COG	Scales and questions on cognition
CRT	Choice reaction time (cognitive test)
MH	Scales and questions on mental health and wellbeing
Other / Administrative	
G2G	Gateway to Global Aging Data

from a (`Stata`) (`.dta`) file, while waves 1–5 were loaded from Statistical Package for the Social Sciences (SPSS) (`.sav`) files.

2. **Automated variable profiling (Python):** For every variable, summary statistics were computed programmatically, including data type, proportion of missing values, number of unique values, most frequent value, and frequency of the most frequent value. These summaries were exported to spreadsheet format for inspection.

3. **Manual variable screening (spreadsheet based):** Using predefined rule based thresholds described below. Variables were manually reviewed and flagged for exclusion based on missingness, lack of variation, dominance of a single category, or administrative function.

4. **Creation of retained variable lists:** Variables not flagged for exclusion were programmatically concatenated in `.xlsx` and manually transferred into wave specific JavaScript Object Notation (JSON) files representing the list of retained *variables of interest*.

5. **Reloading filtered datasets:** The JSON files were then used to subset the original raw datasets, producing reduced datasets with identical rows but fewer column variables. No scripts were used to clean, recode, or impute data at this stage.

Manual filtering followed the criteria below, also visualised in Figure 3.1:

- **Missingness:** Variables with $> 80\%$ missing values were dropped. This threshold retains sufficient feature coverage while avoiding the imputation uncertainty that arises when the majority of values are absent.

- **Unique values:** Drop variables with a single observed value, as they contain no discriminative information and cannot contribute to modelling or inference. Variables with low cardinality (2–4 unique values) were retained for review, as they frequently represent meaningful categorical responses (e.g., binary indicators or ordinal scales) relevant to ageing outcomes.

- **Dominant category (top value/frequency):** Variables where a single category (sample value) marked as the `topValue` occurs over $\approx > 90\%$, `topFrequency` of observations were flagged. Such dominance indicates limited variability, reducing analytical utility and predictive value. Variables exceeding this threshold were marked for review or exclusion.
- **Identifiers / administrative fields:** IDs, version tags, and sampling weights were identified using the PMF metadata. Variable names and descriptions were exported from each wave's PMF file into Comma Separated Values (CSV) files (code provided in Appendix A) to systematically document and review variables prior to filtering. These exports enabled structured manual inspection and traceability between raw PMF metadata and downstream variable selection decisions.

During this process, variables exceeding these exclusion thresholds were visually flagged (e.g., marked in red). Non-flagged variables were concatenated using a spreadsheet formula that wrapped each variable name in double quotes and separated them with commas [1], producing the correctly formatted input for the wave specific JSON files. These files represent the *variables of interest* retained to serve as inputs for the upcoming automated data preprocessing and cleaning.

This process yielded a significantly reduced dataset for each wave provided in Table 3.3. For instance, in wave 1, manual filtering reduced the feature space by 50%. When considering all five waves collectively, the total number of raw variables was reduced from $5,345 \rightarrow 3,113$ representing $2,232$ dropped features with an overall reduction average of $\approx 42.0\%$. All variable exclusion decisions were completed at this stage. Subsequent steps focus on numerical standardisation and model preparation.

3.5 Dataset characteristics after filtering

To further understand the characteristics of these retained variables, the datasets were profiled using an exploratory visualiser (`.ipynb`) file implemented with

[1] The formula used was `=TEXTJOIN(", ", TRUE, CHAR(34) & J2:Jn & CHAR(34))`, applied in Microsoft Excel to each wave's variable list.

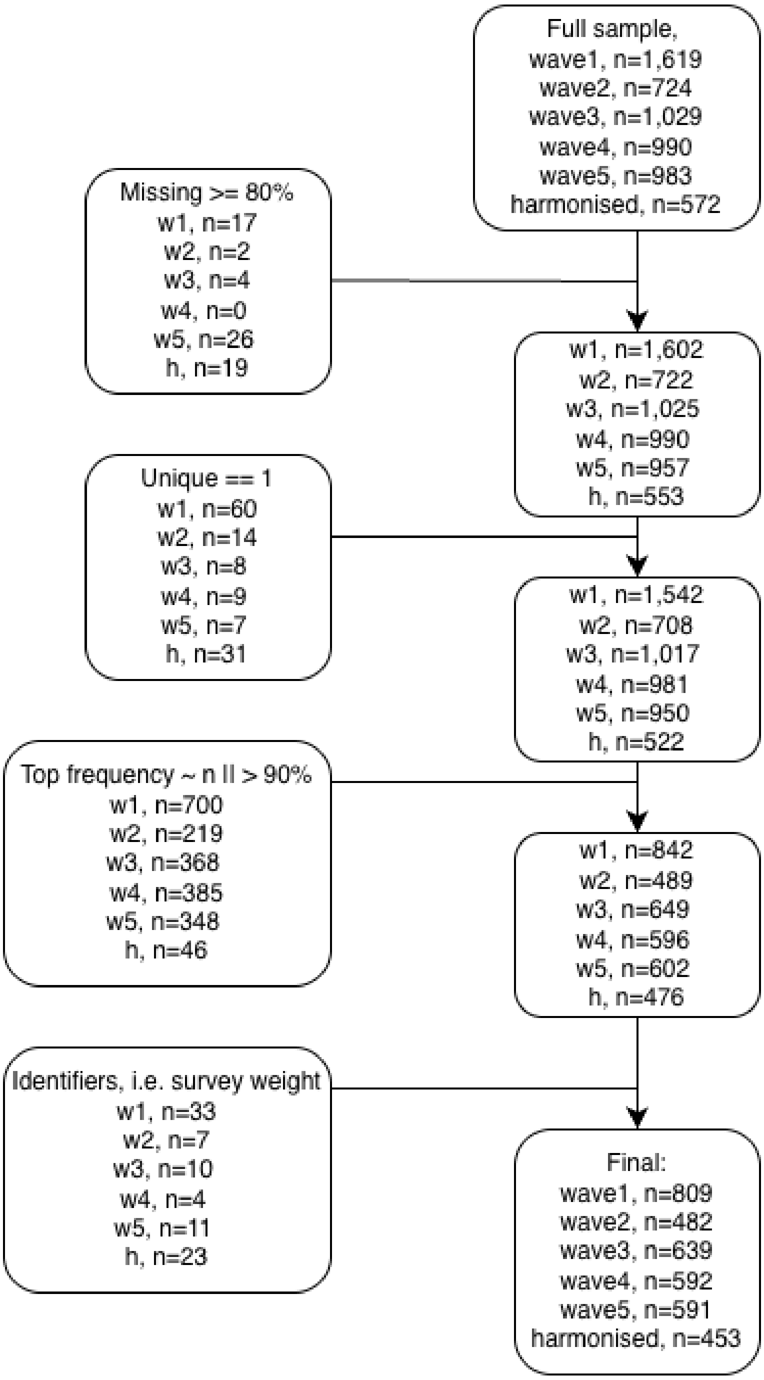

Figure 3.1: Manual workflow for variable screening and filtering. Placeholder (n) values indicate variable counts at each stage.

`pandas`, `numpy`, `matplotlib`, and `seaborn`. This step examined data types, missingness patterns, unique value distributions, and representative sample values across all waves. The purpose of this profiling stage was not transformation but to expose structural inconsistencies and latent formatting issues that would undermine direct numerical conversion.

Table 3.3: Input shape for cleaning pipeline (post manual filtering workflow).

Dataset	Row (N participants)	Column (N Variables)	% Reduction
Wave 1	8501	809	50.0%
Wave 2	7206	482	33.4%
Wave 3	6397	639	37.9%
Wave 4	5713	592	40.2%
Wave 5	4978	591	39.9%
Harmonised	8501	453	20.8%

For example, in wave 1, the visualiser detected 650 categorical columns, 150 (`float64`) columns, and 9 object columns, while the harmonised dataset contained 304 categorical columns, 148 `float32` columns, and several integer columns. Age distributions confirmed that the study population was restricted to older adults across waves, however, ages were bottom coded for participants younger than 50 years and top coded for those aged 65–85 years and above. Missing value summaries indicated substantial variation (e.g., mean missing values ranging from 4.7% in wave 4 to 35% in the harmonised dataset, with some variables exceeding 77% missing).

These insights exposed structural inconsistencies, for instance, variables that appeared numeric in fact encoded labels such as "1.Rarely or none of the time", and many columns contained sentinel codes (e.g., −98, _99). These patterns directly informed the design of the deterministic cleaning pipeline.

3.6 Variable coverage by thematic domain

Using the PMF prefixes described in Table 3.2 as a guide, variables were assigned to seven thematic groups in Table 3.4, each capturing a conceptually coherent domain spanning physical, clinical, cognitive, social, and behavioural aspects of ageing. The heatmap at the base of Table 3.4 shows domain coverage across waves, darker shading indicates a higher variable count highlighting which domains contribute most features to the dataset.

Table 3.4: Thematic domain grouping of longitudinal features.

ID	Thematic Group	Description	Representative Variable Examples	Total Variables
1	**Socio-Demographic & Context**	Baseline demographic and socioeconomic indicators	Age/Sex Education Income Marital status	844
2	**Physical Health & Clinical**	Clinical and self reported health measures	Blood pressure BMI Chronic disease status Self-rated health	853
3	**Functional Status & Frailty**	Physical functioning and frailty markers	ADL/IADL Grip strength Gait speed	174
4	**Lifestyle & Behavioural**	Health behaviours and activity patterns	Physical activity Smoking Alcohol use	120
5	**Nutrition & Metabolic**	Metabolic and biomarker indicators	C-Reactive protein Creatinine Lipid markers	22
6	**Psychological & Cognitive**	Cognitive performance and mental health	Mini-Mental State Examination (MMSE) Depressive symptoms Stress scale	291
7	**Other / Administrative**	Survey metadata and structural variables	Sampling weight Completion status Identifiers (ID)	809

	SocioDemographic_Context	PhysicalHealth_Clinical	FunctionalStatus_Frailty	Lifestyle_Behavioural	Nutrition_Metabolic	Psychological_Cognitive	Other_Administrative	Total_Variables
wave1	272	149	48	58	10	97	175	809
wave2	119	148	18	21	0	38	138	482
wave3	164	184	44	14	8	93	132	639
wave4	161	186	30	13	4	36	162	592
wave5	128	186	34	14	0	27	202	591
Total_Per_Domain	844	853	174	120	22	291	809	3113

Thematic domain dominance across waves (higher n features per domain).

Chapter 4

Technical Implementation

4.1 Automated cleaning engine

The core challenge in converting heterogeneous variable types across the six datasets was achieving a single numerical standard suitable for downstream analysis. This required enforcing *numerical homogeneity*, defined here as representing all retained variables using a single numerical representation (`float64`) with a unified missing or non response sentinel (`-1.0`).

Categorical variables were dominant (2,916 columns) while numeric types were fragmented, with `float64` (462 columns) adopted as the common standard to support decimal values from midpoints, ordinal scaling, and frequency normalisation. The solution was implemented as two stages: automated *variable scenario detection* function (Appendix C) and a deterministic *six step numerical cleaning engine* (Appendix E). Nine scenario categories were defined initially and three served only as validation fallbacks and were eliminated as coverage improved, leaving six active transformation steps applied uniformly across all waves (Appendix B).

4.1.1 Stage 1: Variable scenario mapping

All variables passed sequentially through the six step pipeline, most were resolved in steps 1–5, with step 6 (Appendix D) handling only residual variables that could not be resolved by earlier transformations.

To ensure robustness, the classification output was immediately processed by a validation routine that generated a comprehensive dictionary of variable states before transformation and feature representations after, including all representative sample values for each variable. This allowed direct inspection of how the same variable was represented across waves prior to cleaning, particularly in cases where identical semantic categories appeared in different orders or formats. By comparing these grouped samples before and after each cleaning step, classification errors and inconsistent mappings were identified and corrected prior to execution of subsequent transformations.

The dictionary generated inside the `show_grouped_vars_for_scenarios` function (Appendix F) was formatted to create an automatic friendly Markdown file for rapid, surface level verification, which honed the final mapping dictionary (an example is provided in Appendix G) for all features across all waves. Automated detection followed by manual validation identified and corrected classification errors before step transformations were executed.

4.1.2 Stage 2: Deterministic six step cleaning sequence

The nine input scenarios were processed sequentially by the *six step cleaning engine*, resolving simple and unambiguous cases first to maximise the proportion of successfully transformed variables before tackling more complex structures, particularly censored and multi-class scenarios (Figure 4.1). With later steps progressively shifting from cleaning into semantic feature engineering, this deterministic sequence applied identical transformation and feature encoding logic to each wave. Particularly steps 4–6, performing rule based feature engineering by deriving semantically meaningful numerical representations directly from observed category labels at runtime. The rationale for this embedded feature engineering approach is detailed in Section 4.1.3, implementation is provided in Appendix D. This design reflects the empirical diversity of survey encodings. A small subset of variables only reveals its structure once simpler transformations have been applied.

The final cleaning pipeline consisted of six deterministic steps applied uniformly across all TILDA waves, with increasingly complex transformations han-

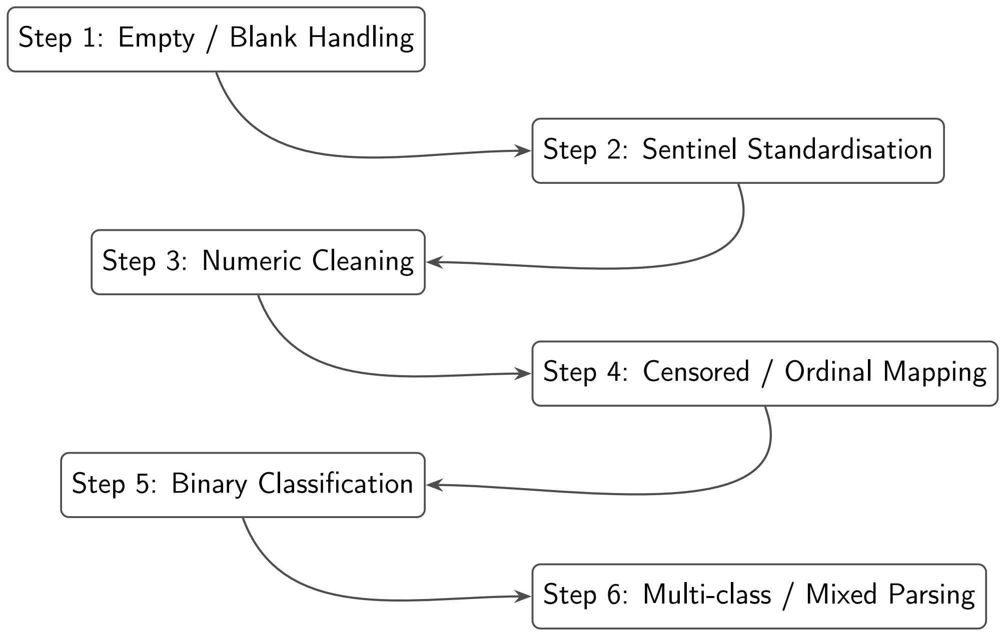

Figure 4.1: Six step deterministic cleaning pipeline applied uniformly across all TILDA waves. Transformation tables and implementation scripts are provided in Appendices B.7 and E.

dled in later stages. Steps 4–6 required particular attention due to hybrid, ordinal, or multi-class formats. A summary of each step and examples is provided below.

1. **Empty / blank values:** All missing entries, empty strings, and whitespace were converted to the standard non-response sentinel (`-1.0`).

2. **Sentinel standardisation:** Wave specific non-response codes (e.g., `-98`, `99`) and textual sentinels (e.g., "`dk`", "`not applicable`") were mapped to the unified (`-1.0`) placeholder.

3. **Numeric transformation:** Numeric like columns containing embedded units, symbols, or hybrid strings (e.g., `50+`, `<10`, `1−2.5`) were parsed using regex and converted to `float64`. For ranges, midpoints and/or boundary values were used. For censored or annotated values, appropriate upper/lower bounds were applied.

4. **Censored / ordinal variables:** This step addressed variables representing

ordinal scales, bracketed numeric ranges, or frequency expressions (e.g., "2–3 days a week", "daily"). Transformations included:

- Converting textual ordinal labels and Likert scale responses to numeric equivalents.
- Mapping calendar based frequency descriptors to numeric weekly equivalents.
- Manual verification was conducted for low count or unusual variables using the Markdown audit outputs to confirm semantic meaning was preserved.

5. **Binary categorical variables:** Two level features (e.g., Yes/No, Male/Female) were dynamically encoded using mappings derived from the observed value set of each variable:

   - Semantic rules were applied e.g., ("no", "not", "female" → 0.0) and ("yes", "male" → 1.0).
   - Count based detection verified that variables had exactly two non-missing unique values.
   - Features with more than two sample values were also analysed for sentinel presence:
     - If a variable had three observed values and one was a sentinel, it was treated as binary.
     - If a variable had four observed values and two were sentinels, it was treated as binary.
   - Sentinels were preserved as `-1.0`.
   - This approach ensured order invariant encoding across waves, even when original labels or numeric prefixes differed.
   - *Example (sentinel case):* Feature `X` with pre-transformed values `['NOT Getting in or out of bed', -1.0, , 'Getting in or out of bed']` was encoded as `[0.0, -1.0, -1.0, 1.0]`, demonstrating how a sentinel preserves binary semantics.

6. **Multi-class / mixed variables (rule based feature engineering):** Variables with remaining multi-class structure, mixed types, or unresolved string numeric hybrids after earlier transformations were processed in the final step. The implementation script is provided in Appendix D:

   - Text based parsing extracted numbers, ranges, units, percentages, and frequency descriptors into numeric values, yielding interpretable numerical feature representations.
   - A word-to-number mapping resolved textual numerals (e.g., `‘twice’`, `‘a couple’`) to numeric equivalents, and a curated semantic expression map handled survey specific phrasing (e.g., `‘not at all concerned’`, `‘daily / almost daily’`) that couldn't be resolved by ordinal position alone.
   - Original values were grouped and assigned numeric identifiers in **first seen order**, non numeric categorical labels were assigned integer codes above the highest observed numeric value to prevent collisions.
   - Manual inspection using the Markdown audit file verified that semantic equivalence was maintained across waves.
   - Step 6 was designed to be *order invariant*, dynamically interpreting category values within each wave rather than relying on predefined lookup tables shared across waves. This preserved consistent numeric representation regardless of variable ordering or wave specific encodings.
     - *Example:* Step 5 handled simple binary variables, e.g., `[‘pain’, ‘No pain’]`$\rightarrow$`[1.0, 0.0]`. In contrast, step 6 dynamically processed multi-class variables, e.g., (pre tax wage/salary) many of which contained over 20 distinct pre transformed categories (e.g., `-1.0, ‘200-299’, ‘Less than 100’, ..., ‘6,000-6,999’`) into consistent numeric representations (e.g., `-1.0, 249.5, 99, ..., 6499.5`). This illustrates why step 6 required wave specific, dynamic parsing to maintain semantic equivalence across diverse observed values.

4.1.3 Embedded feature engineering

Although the six step pipeline is framed primarily as a numerical cleaning procedure, steps 4–6 also perform explicit rule based feature engineering by interpreting heterogeneous categorical survey responses and converting them into semantically meaningful numerical representations. Rather than acting as a separate post processing stage, feature engineering is embedded directly within the cleaning logic, formalising ordinal structure, frequency, and magnitude encoded in text into directly modelable `float64` values suitable for downstream analysis.

Crucially, this process engineers the representation of each feature rather than expanding the feature space. Free text frequency responses such as "`once or twice a week`", "`a couple of times per month`", or "`almost every day`" are not treated as nominal categories. Instead, lexical cues (e.g., `once`, `twice`, `couple`) are resolved to numeric quantities and normalised onto a common temporal scale, yielding a continuous rate representation. Implicit semantic information such as ordering, intensity, and frequency is therefore preserved through deterministic parsing and normalisation without introducing additional variables. This design maintains the original dimensionality of the dataset while ensuring numerical homogeneity and semantic integrity, producing a compact and model ready representation of complex real world survey data.

4.1.4 Cross wave semantic verification

Given the absence of guaranteed category ordering across waves, preserving *semantic equivalence* during numerical encoding and feature representation was essential to avoid introducing distortions into downstream ML modelling. Pre and post transformation values for each group were exported to a readable Markdown (`.md`) file. This representation confirmed that features of known importance for downstream predictive models were preserved and correctly encoded. Self-rated general health, comparative self-rated health, and smoking frequency are among the most influential predictors in FI based mortality models [6]. Retaining these features from the manual filtering thresholds applied in Section 3.4 and encoding them consistently across waves ensured that the cleaned dataset maintains coverage and semantic integrity, preventing distortions that could potentially compromise model performance.

A key challenge in cleaning the five TILDA waves was the lack of guaranteed ordering for categorical values of the same variable across different waves. Without semantic equivalence, the same health response (e.g., "Good" health) encoded as category 1 at wave 4 but category 4 at wave 5 would be treated as different constructs by any model. This is particularly important for features like self-rated health, where consistent ordinal meaning across all five waves is essential for meaningful prediction. Table 4.1 illustrates this encoding risk for waves 4 and 5.

Gender (`sex`) variables appeared under different value encodings, with varied category orderings and numeric prefixes across waves. Values such as `['Male', 'Female']`, `['Female', 'Male']`, and prefixed forms like `["1.male', '2.female']` were all mapped to a consistent numeric representation (`1.0` for male, `0.0` for female). This ensured that semantic equivalence, rather than positional ordering or wave specific encoding, determined numerical assignment.

The Markdown audit confirmed that semantically identical categories were correctly grouped across waves regardless of variable name or encoding order, and that complex value spaces (e.g., income bands, frequency descriptors) were being collapsed to stable numeric representations consistently across waves. Full audit wave examples are provided in Appendix G.

4.2 Validation of the transformed features

Validation checked whether known biological relationships reported in the literature were preserved after cleaning. Due to restricted access to certain outcome variables within TILDA, full end-to-end replication of all published models was not always feasible. In particular, access through the hotdesk facility for the official mortality outcomes is subject to additional governance requirements, including institutional GDPR training certification and approval by the TILDA data access committee [42]. As a result, validation focused on two complementary strategies:

Table 4.1: Inconsistent category ordering for comparative self-rated health ("How does your health compare to others your age?") across waves and its impact on positional label encoding.

Pos	Wave 4		Wave 5	
	Category	**Code**	**Category**	**Code**
1	Good	3.0	Very Good	4.0
2	Fair	2.0	Fair	2.0
3	Poor	1.0	Excellent	5.0
4	Excellent	5.0	Good	3.0
5	DK	-1.0	Poor	1.0
6	Very Good	4.0	DK	-1.0
7	RF	-1.0	RF	-1.0

(i) Internal consistency checks against established biological and clinical relationships, and

(ii) Constrained replication of published modelling frameworks using available predictors and the derived 8 year mortality proxy outcome.

These validation steps help confirm that the cleaning pipeline retains expected relational patterns and produces datasets suitable for the modelling tasks.

4.2.1 Internal feature relationships

The first validation stage assessed whether cleaned predictors preserved biologically and clinically plausible relationships reported in prior TILDA studies. This analysis focused on predictors identified by [4] as accounting for approximately 95% of the explained variance in Gait Speed Reserve (GSR), defined as the difference between maximum and UGS.

As GSR is not directly available in the public wave 3 dataset, validation was performed by examining the directional relationships between the 7 primary predictors themselves. This approach evaluates whether the encoding process, including sentinel handling, missing value treatment, and ordinal mapping, preserved the expected structure of the data rather than simply fitting to a particular outcome.

Figure 4.2 presents the pairwise Pearson correlation matrix for the selected predictors after the deterministic cleaning pipeline. GRIP exhibited a negative association with both age and chair stand time, reflecting the inverse relationship between muscular strength and physical performance decline. Cognitive performance (Montreal Cognitive Assessment (MOCA)) demonstrated a moderate negative correlation with age and a positive association with educational level, consistent with known patterns of cognitive reserve in older adults. Education correlated positively with cognitive performance, and BMI showed the expected weak negative association with age, reflecting reduced physiological reserve in older cohorts.

Table 4.2 summarises the expected and observed directional relationships between key predictor pairs. Notably, GRIP showed moderate negative correlations with both age ($r = -0.265$) and chair stand time ($r = -0.170$), reflecting the expected inverse relationship between muscular strength and physical performance decline. Cognitive performance (MOCA) displayed a moderate negative association with age ($r = -0.381$) and a positive association with education ($r = 0.367$), consistent with known patterns of cognitive reserve in older adults. BMI demonstrated the anticipated weak negative correlation with age ($r = -0.073$). These results confirm that the cleaning pipeline retained the expected numerical structure and relational patterns of the predictors, rather than introducing artificial associations.

4.2.2 Model based validation

To evaluate whether the cleaned dataset preserves predictive utility, a constrained replication of [6] was conducted using available wave 1 predictors. The study predicted 8 year mortality using 32 FI items. In the cleaned wave 1 dataset following the cleaning pipeline described in Section 4.1.2, only 16 of these 32 FI items remained following the manual variable filtering described in Section 3.4, which prioritised global numeric consistency across waves.

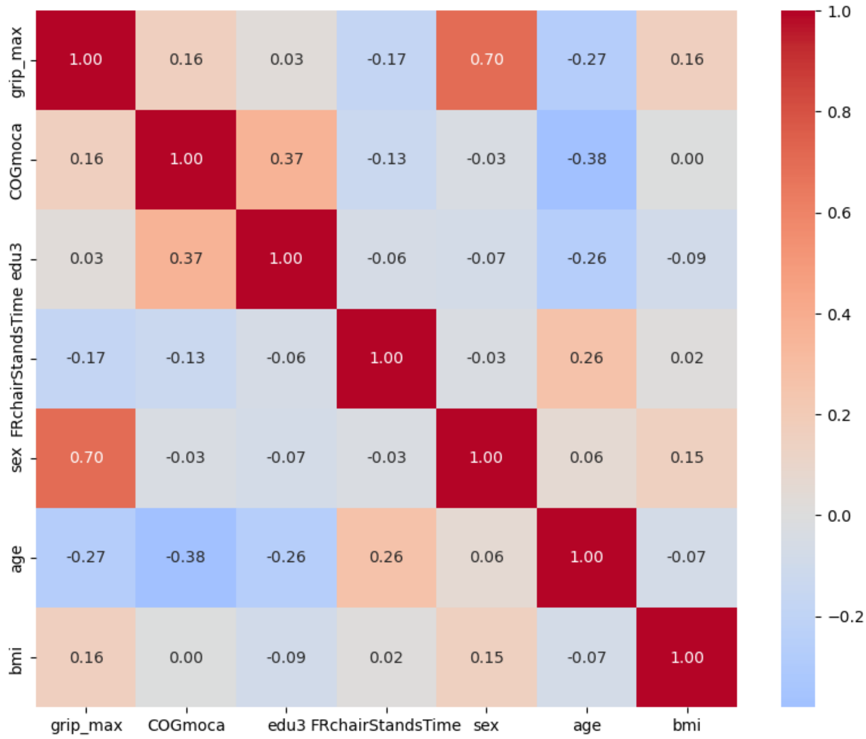

Figure 4.2: Pairwise Pearson correlations for the seven predictors used in directional validation (wave 3).

Table 4.2: Directional validation of the seven features accounting for 95% of gait speed reserve variance as identified by [4].

Predictor 1	**Predictor 2**	**Expected Direction / Justification**	**Observed r**
GRIP	Age	**Negative (−)**: GRIP is a positive predictor of GSR, while age is a negative predictor.	-0.265
GRIP	Chair Stand Time	**Negative (−)**: Higher strength (grip) is expected to correlate with faster (lower) performance time in physical tests.	-0.170
Montreal Cognitive Assessment (MOCA) Score	Age	**Negative (−)**: Cognitive performance (MOCA) contributes positively to GSR, whereas increasing age is associated with reduced reserve.	-0.381
Education	MOCA Score	**Positive (+)**: Third level education and cognitive scores are both cited as positive contributors to cognitive reserve and GSR.	0.367
Body Mass Index (BMI)	Age	**Weak / Mixed**: BMI and age both negatively impact GSR, though their internal correlation in older cohorts is typically weak.	-0.073

Due to access restrictions, the official mortality data were unavailable. As a result, a *8 year mortality proxy* outcome was constructed to approximate the sex specific 8 year mortality rates reported in [6] (men: 14.9%, women: 11.1%), derived from linkage to Ireland's Central Register Office records [43]. Mortality status was first assigned by randomly selecting deceased participants from the wave 1 cohort aged 50 years or older to match the published sex specific mortality proportions observed at wave 5 follow up. An equal number of surviving participants were then randomly sampled within each sex to construct balanced datasets for model training. The resulting random proxy datasets included 1,114 men (557 deceased) and 982 women (491 deceased), reflecting the baseline wave 1 population aged $\geq$ 50 years ($n = 8,172$: 3,743 men and 4,429 women). Two variants of this proxy were implemented:

(i) the purely random assignment described above, used as a structural baseline, and

(ii) biologically informed probabilistic assignment based on latent FR structure extracted via Principal Component Analysis (PCA). This dual construction allowed separation of statistical artefact from biologically meaningful signal.

The random proxy serves as an initial diagnostic benchmark. Mortality labels were assigned by randomly selecting deceased participants within each sex to match the published 8 year mortality proportions. While this preserves overall class balance, it does not encode any biological or FR related information and therefore serves only as a structural validation baseline.

For the biologically informed proxy, the 16 available FI items were standardised to zero mean and unit variance using z-score transformation. PCA was then performed separately within each sex on complete case observations to extract the principal component representing the dominant latent FR structure. The First Principal Component (PC1) was retained as a latent FR score, and its direction was aligned such that higher PC1 scores correspond to higher FR by enforcing a positive correlation with the calculated FI. This latent FR score was subsequently combined with age to construct a probabilistic mortality assignment reflecting both biological vulnerability and time dependent risk. Two model variants were tested for each sex:

1. **FI only model:** using the 16 available FI items, without age.

2. **FI + Age model:** including age as an additional predictor.

Model training and evaluation followed the procedure reported in [6] as closely as permitted by the available data. Features were min-max normalised prior to modelling. For each sex, data were split into training (80%) and testing (20%) sets. To mitigate overfitting and obtain robust estimates of model performance, 5-fold stratified CV was applied on the training data. AUC was selected as the primary evaluation metric due to its widespread use in prior TILDA and ML based studies (Figure 2.2) and its robustness to class imbalance.

For six of the sixteen FI items, the cleaned dataset transformations differed from the scoring rules published in [6]. To align precisely with the original 16/32-item FI methodology, these items were recoded following the published ordinal and dichotomous mappings. This step was necessary to ensure that the six remapped FI items reflects the same interpretation of health deficits used in the literature and is consistent with standard FI construction principles [16]. Models were then re-evaluated using this partial remapping FI to assess sensitivity to feature encoding.

The resulting AUC values are reported separately for (i) originally cleaned (Figure 4.3) vs 6/16 recoded FI items (Figure 4.4), (ii) random vs PCA informed mortality assignment, (iii) FI only vs FI + Age models, and (iv) men and women. These results highlight several key points:

1. **Random proxy baseline (cleaned vs recoded FI):** Under purely random mortality assignment, discrimination remained at chance level regardless of FI version. The cleaned 16 FI items achieved AUC = 0.507 (men) and 0.512 (women). Recoding six items per the published scoring rules [6] improved this to 0.567 (men) and 0.537 (women), but discrimination remained close to chance. This confirms that without a biological structure in the outcome labels, predictive performance reflects noise.

2. **PCA informed mortality proxy (cleaned vs recoded FI):** When mortality assignment incorporated the PCA derived FR representation PC1,

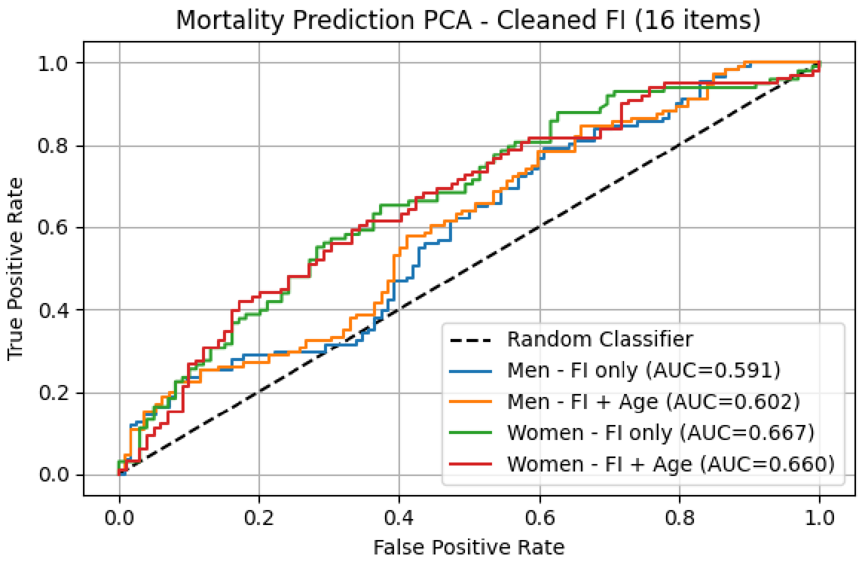

Figure 4.3: Receiver Operating Characteristic (ROC) curves for mortality prediction using 16 cleaned frailty index items under the PCA informed mortality proxy. Separate models are shown for men and women, with and without age.

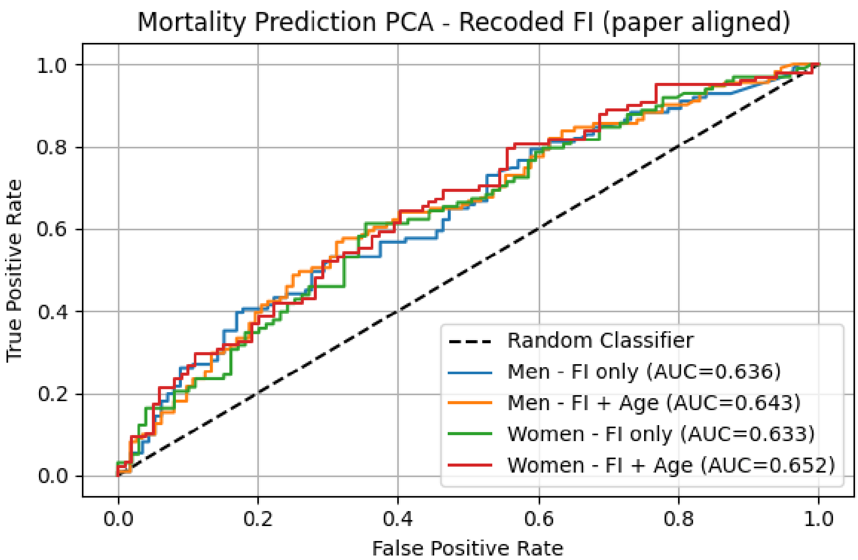

Figure 4.4: ROC curves for mortality prediction using the 6/16 recoded frailty index items aligned with [6]. Performance is shown separately for men and women, with and without age.

discrimination improved relative to the random baseline for both FI constructions. For the 16 cleaned FI items, the FI only model achieved AUC = 0.591 (men) and 0.667 (women), increasing to 0.602 (men) and 0.660 (women) when age was included. When six of the sixteen deficits were recoded in accordance with the published scoring framework, the FI-only model achieved AUC = 0.636 (men) and 0.633 (women), while the FI + age model achieved 0.643 (men) and 0.652 (women).

This indicates that closer alignment with published ordinal deficit scoring improves discrimination relative to the simplified numeric encoding applied during initial cleaning.

3. **Comparison with published performance:** Absolute AUC values remain below those in [6] (approximately 0.70 for FI only, up to 0.80–0.90 with age), as expected given the created proxy outcome and only 16 of the original 32 FI items. The methodological sequence is preserved. Random assignment gives chance performance, PCA latent FR structure improves discrimination, and adding age further improves prediction.

   (i) Random assignment yields chance level performance,

   (ii) Extracting latent FR structure via PCA improves discrimination,

   (iii) Inclusion of age generally improved mortality prediction.

   The cleaned dataset maintains realistic sex distributions (45.8% men, 54.2% women) and mortality proportions consistent with published estimates, confirming that preprocessing preserved demographic and structural integrity required for downstream modelling.

Chapter 5

Empirical Evaluation

5.1 Modelling and evaluation framework

Building on the cleaned and validated datasets, this chapter presents the empirical evaluation of ML models for predicting chronic disease onset in the TILDA cohort. The evaluation is based on participants aged between 50–84 years with a mean age of 66. To ensure that the prediction task reflects new onset disease, individuals with the target condition already present at the start of a transition were excluded, leaving only at risk observations for training and testing. This rule was applied consistently across all designs and for both Track A (*any disease outcome*) and Track B (*disease specific outcomes*).

Final analysis samples differed by design and tracks due to data structure. The baseline design (D1) uses a single transition (Wave 1→Wave 5), with a follow-up period of approximately 8 years. The pooled multi-wave designs (D2 and D3) aggregate all eligible consecutive transitions (W1→W2, W2→W3, W3→W4, W4→W5), each spanning approximately 2 years, thereby capturing shorter term dynamics.

Sample sizes and incident rates for Track A are summarised in Table 5.1. Track B cohort sizes and incident rates vary by condition. Appendix H provides these details for each disease specific model.

5.1.1 Outcome definitions

Two complementary prediction tasks were defined.

Track A–Predicting any chronic disease onset: Track A uses a binary outcome indicating the first occurrence of any chronic disease during follow-up.

- In D1, disease free status at wave one was established using a composite of self-reported conditions (e.g., eye conditions, cardiovascular conditions, diabetes related conditions, general chronic conditions, limiting long term illness, and self-reported chronic/health problems). Incident cases were then defined using any positive ICD-10 disease codes from the WHO (*International Statistical Classification of Diseases and Related Health Problems 10th Revision*) at wave five (e.g., mental and behavioural disorders & musculoskeletal diseases).
  - No composite ICD-10 indicator exists in wave 1 to establish a clinical baseline. These clinical codes were recorded unevenly across the middle waves, with only three indicators available in wave 2, two in wave 3, and one in wave 4, before a comprehensive set of 16 variables appeared at wave 5, partially a consequence of the manual low heterogeneity filtering flow described in Section 3.4.
  - Physical & Cognitive Health (PH) variables reflect prior doctor diagnoses ("has a doctor ever told you"), whereas the "self-reported long term chronic/health problem" variable captures general perceived health burden without requiring clinical confirmation.
  - All variable scoring schemes and labels are provided in Appendix I.
- In D2 and D3, the "self-reported long term chronic/health problem" variable (where $y = 0$ denotes no chronic illness and $y = 1$ denotes presence of chronic illness) was used to both define the outcome and identify the at risk cohort (those with $y = 0$ at transition start) for inclusion in each wave. This variable is available across all waves, enabling consistent application

across all four consecutive transitions.

$$\text{Coverage}_y = \frac{\sum_{i=1}^{N} \mathbb{1}\big(y_i \neq missing(-1.0)\big)}{N} \times 100 = 99.91\%$$

where N denotes the number of person wave observations and y_i the binary outcome for observation i, hence Coverage_y reports the percentage of observations for which y is recorded (i.e., not coded as the missing value -1.0) for the self-reported long term chronic/health problem outcome.

- An analysis of wave 5 data compared self-reported long term chronic/health problem against any incident case clinical records (ICD-10). While raw agreement reached 59.5%, *Cohen's Kappa for agreement beyond chance* ($\kappa = 0.167$) indicates poor agreement, showing a substantial discrepancy between participant perception and clinical documentation. Among participants with a confirmed clinical diagnosis, 45% reported themselves as healthy, whereas 38% of those without a clinical diagnosis reported a long term chronic/health problem. This low agreement suggests that "self-reported long term chronic/health problem" and clinically coded disease are distinct but complementary constructs, one capturing perceived illness burden and the other capturing objectively documented diagnoses.

Table 5.1: Sample sizes and incidence rates by design for Track A.

Design	**Observations (n)**	**Incident cases (n)**	**Incidence rate (%)**
D1: Baseline (W1→W5)	916	248	27.07
D2: Multi-wave pooled	14,296	3,212	22.5
D3: Top-20 (reduced)	14,296	3,212	22.5

Track B–Disease specific prediction: Track B targets individual chronic disease outcomes, e.g., circulatory disease, diabetes, and functional decline. For each prediction task, the outcome was defined as the first recorded diagnosis of the specific condition, and participants were excluded if the condition was present at the baseline of the corresponding transition. Scoring details for each disease specific outcome are also provided in the same Appendix I. Additionally, a predictor add-on analysis was conducted to evaluate whether individual candidate features from domains beyond the top-20 feature set detailed in Appendix J contribute independent predictive value.

5.1.2 Modelling designs

1. **LR** — An interpretable linear baseline consistent with its established use in TILDA mortality and disease prediction [35], providing coefficients and odds ratios directly. In D1 and D2, a fixed configuration was used: `solver="liblinear"`, `C=0.001`, `class_weight="balanced"`, `max_iter=2000`. In D3, hyperparameters were tuned via `RandomizedSearchCV` over `C` (log-uniform $[10^{-4}, 10]$), `solver=["saga"]`, `l1_ratio` (uniform $[0, 1]$), and `max_iter` $\in \{1000, 2000, 5000\}$.

2. **XGB** — A gradient boosted tree ensemble selected for its demonstrated performance in TILDA diabetes prediction using multidomain predictors [25], capturing nonlinear interactions in heterogeneous tabular health data. In D1 and D2, fixed parameters were used: `n_estimators=1500`, `max_depth=3`, `learning_rate=0.03`, `min_child_weight=5`, `gamma=0.1`, `subsample=0.8`, `colsample_bytree=0.7`, with class imbalance handled via `scale_pos_weight` (negative to positive ratio). Early stopping (`early_stopping_rounds=50`) on an internal 20% validation split determined the optimal number of boosting rounds. The model was then refit on the full training set at that iteration. In D3, all tree and regularisation parameters were additionally tuned via `RandomizedSearchCV`.

3. **RF** — A bagging based ensemble with established use in falls, frailty, and functional decline prediction within TILDA [6, 19], providing robustness through random feature subsampling (`max_features="sqrt"`). Fixed parameters in D1/D2: `n_estimators=500`, `max_depth=5`, `min_samples_leaf=10`, with class imbalance handled via `class_weight={0:1, 1:neg/pos ratio}`. In D3, `n_estimators` (100–800), `max_depth` (3–10), `min_samples_leaf` (5–30), and `max_features` $\in$ {`"sqrt"`, `"log2"`} were tuned via random search.

4. **MLP** — A feedforward neural network with two hidden layers, included to represent the neural family in the four model comparison. Neural networks

have been proposed as a promising tool for clinical risk prediction, a capacity noted by [44] in their survey of ML applications for clinical risk prediction. This inclusion here tests directly whether this advantage holds in survey based longitudinal ageing data against the tree-based baselines. Fixed parameters in D1/D2: `hidden_layer_sizes=(32,16)`, `activation=‘‘relu’’`, `solver=‘‘adam’’`, `alpha=0.01`, `learning_rate_init=0.001`, `max_iter=500`, with early stopping (`n_iter_no_change=20`, `validation_fraction=0.1`). Class imbalance was handled via `compute_sample_weight(class_weight= ‘‘balanced’’)` passed at fit time, as MLP does not natively support `class_weight`. In D3, `hidden_layer_sizes` $\in \{(32,), (64,), (32{,}16), (64{,}32), (128{,}64)\}$, `activation` $\in$ {`‘‘relu’’`, `‘‘tanh’’`}, `alpha` (log-uniform $[10^{-4}, 1]$), and `learning_rate_init` (log-uniform $[10^{-4}, 10^{-1}]$) were tuned via random search.

The preprocessing backbone median imputation (`SimpleImputer`), variance filtering (threshold = 0.01), and missingness filter (> 50% on training only) was shared across all models, with scaling (`StandardScaler`) applied only to LR and MLP (sensitive to feature magnitude). Tree-based models were trained on the raw imputed features without scaling. All steps were fit exclusively on training data (prior to internal train/validation splitting) before being applied to validation and test sets, preventing information leakage.

Track B models used XGB exclusively, as it achieved the strongest and most consistent performance across D2 and D3 in Track A (see Section 5.2). The same model configuration and hyperparameter search space used in D3 was applied in Track B. The three primary modelling designs were implemented to address and assess the effect of baseline prediction, longitudinal pooling, and feature reduction with optimisation, respectively. The full pipeline structure across all three designs and both tracks is illustrated in Figure 5.1.

- **D1 (Baseline):** A single W1→W5 transition. Only participants disease free at wave 1 (using self-reported and derived disease variables) were included, with incident disease defined by any positive ICD-10 coded diagnosis at wave 5 ($n = 916$, incidence 27.1%). Feature selection used `SelectKBest` ($k = 20$) and CV used `StratifiedKFold` (one observation per participant).

- **D2 (Multi-wave pooled):** All four eligible consecutive transitions (W1→W2, W2→W3, W3→W4, W4→W5) were stacked into a single pooled dataset. Disease free status and incident outcome were both defined using the self-reported long term chronic/health problem variable at each transition. A numeric wave transition indicator was added as a feature to capture temporal context. The pooled cohort comprised of $n = 14{,}296$ person wave observations (incidence 22.5%). Feature selection used `SelectKBest` ($k = 100$), and CV used `StratifiedGroupKFold` to ensure no participant appeared in more than one fold.

- **D3 (Top-20 + Hyperparameter Optimization (HPO)):** Built directly on the D2 cohort and wave structure. The feature set was restricted to the 20 highest ranked predictors by XGB gain importance from D2, plus the wave transition indicator. No additional feature selection step was applied. HPO was performed for all four model families using `RandomizedSearchCV` (200 iterations) within a `StratifiedGroupKFold` framework.

The choice of k was driven by sample size. D1 ($n = 916$) is a small cohort, restricting to $k = 20$ avoids overfitting and gave the most stable cross-validated performance. The D2 pooled cohort is approximately 15 times larger and can support a proportionally larger feature set, $k = 100$ produced the strongest results at that scale without signs of overfitting. Applying those same 20 features to the full D2 cohort matched D2 level performance within $\Delta\text{AUC} = 0.002$ in D3, confirming that the cross-design gain is driven by longitudinal structure rather than feature count.

All three designs shared the same participant level train/test split (80%/20%, stratified by outcome), the same missingness filter and variance threshold applied on training data only, and the same five fold CV evaluation protocol. Track B models were evaluated under both the D2 (all X) and D3 (top-20 X) structures for each disease specific outcome. Full details are provided in Subsection 5.2.4.

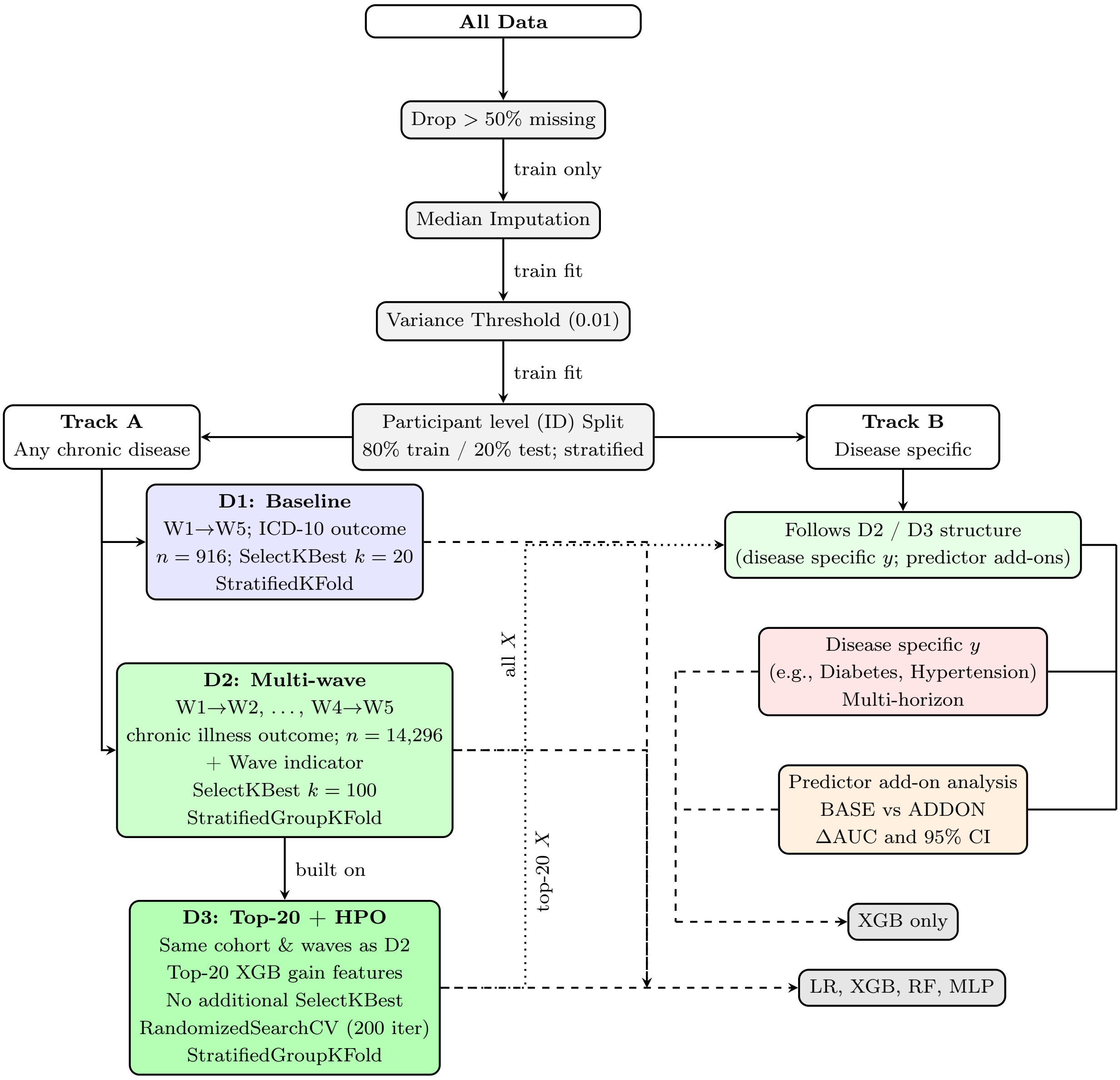

Figure 5.1: Pipeline overview: shared preprocessing applied to all data before branching into Track A (any chronic disease, D1/D2/D3) and Track B (disease specific, following D2/D3 structure). D3 is built directly on D2's cohort and wave structure, adding feature reduction and hyperparameter optimisation.

Figure 5.1 note: Feature count after the missingness filter→D1 retains ≈570 features, further reduced to $k = 20$ by `SelectKBest`. D2 retains ≈261 features, reduced to $k = 100$. D3 uses exactly 21 features (top-20 + wave indicator) with no further selection.

5.1.3 Participant level data splitting and leakage prevention

All splitting was performed at the **participant level**. Each participant was allocated to exactly one of the three data partitions, training, validation, or test set, so that no individual contributed observations to more than one split. The overall dataset was first split into 80% for model development and 20% as a held-out test set. The 80% development part was then further divided 80%/20% internally, yielding a final three way allocation of 64% training, 16% validation, and 20% test. Splitting was performed using participant identifiers to ensure group wise separation, with stratification by outcome to preserve class balance across splits. As demonstrated in longitudinal brain Magnetic Resonance Imaging (MRI) analysis [45], participant level splitting prevents identity confounding, where models learn individual health signatures rather than generalisable risk patterns.

D1 used `StratifiedKFold` ($k = 5$). D2 and D3 used `StratifiedGroupKFold` ($k = 5$), ensuring all observations from a given participant are assigned to a single fold and no participant appear in both training and validation sets simultaneously. Three additional leakage controls were applied across all designs:

- **Pipeline isolation:** Imputation medians, variance masks, and feature selection scores were computed on the training set only. The resulting transformations were then applied unchanged to the validation and test splits.
- **Temporal ordering:** Predictors were drawn from the baseline wave of each transition (wave n) to predict the next wave outcome (wave $n + 1$), preventing look ahead bias.
- **Exclusion of direct proxies:** Features that directly encode the outcome being predicted (e.g., self-reported disease source indicators that feed into the ICD-10 codes used as outcome, such as ("has a doctor ever told you

that you have") a Ministroke or TIA and Parkinson's disease for nervous system diseases), were removed from the predictor set prior to modelling.

5.1.4 Evaluation protocol and metrics

Model performance was assessed using discrimination, calibration, and classification metrics. The primary metric was AUC. Models requiring HPO (D3) used `RandomizedSearchCV` within the same CV framework.

Calibration was evaluated using the Brier score, which measures the accuracy of predicted probabilities, while classification performance was summarised using the F1-score (harmonic mean of precision and recall) (F1) at the optimal decision threshold, capturing the balance between precision and recall in the presence of class imbalance (e.g., $22 - 27\%$ incidence). Additional metrics, including sensitivity, specificity, and confusion matrices, were used to support interpretation.

For Track B predictor add-on analysis, 95% Confidence Interval (CI)s for AUC and Average Precision (AP) deltas were estimated via bootstrap resampling ($B = 2{,}000$ resamples) to ensure statistical significance.

5.2 Main results

Performance is evaluated first across the three Track A designs (D1, D2, and D3), followed by cross design and model comparisons. Results conclude with Track B disease specific outcomes.

5.2.1 Track A: Any disease onset

Results are presented across the three modelling designs, with tables summarising discrimination (AUC), calibration (Brier score), and classification performance (F1), alongside training, validation, and test metrics to assess generalisation.

5.2.1.1 Design 1 performance

Table 5.2 summarises the performance of the four models on the single transition (Wave 1 $\rightarrow$ Wave 5). Discrimination is weak with test AUC values ranging from

0.53–0.58, and the larger train-test gaps for tree-based models indicate stronger overfitting in the baseline design.

Table 5.2: Predictive performance across model families for D1 (baseline design).

Model	CV AUC	Train AUC	Val AUC	Test AUC	Brier Score	F1 Optimal
Logistic Regression	0.589	0.677	0.582	0.571	0.245	0.411
XGBoost	0.569	0.825	0.563	0.544	0.240	0.398
Random Forest	0.579	0.871	0.565	0.534	0.231	0.440
MLP	0.536	0.664	0.552	0.582	0.241	0.419

5.2.1.2 Design 2 performance

Table 5.3 shows the results when using all consecutive transitions (pooled multi-wave design). All models improve substantially compared to D1, with test AUC values ranging from 0.69 to 0.72.

Table 5.3: Predictive performance across model families for D2 (multi-wave design).

Model	CV AUC	Train AUC	Val AUC	Test AUC	Brier Score	F1 Optimal
Logistic Regression	0.715	0.728	0.715	0.709	0.217	0.460
XGBoost	0.716	0.736	0.719	0.715	0.214	0.468
Random Forest	0.711	0.740	0.717	0.711	0.212	0.452
MLP	0.698	0.749	0.706	0.699	0.208	0.455

5.2.1.3 Design 3 performance

Table 5.4 presents performance using only the 20 most informative predictors (derived from XGB model trained on D2) with HPO applied across all models. Discrimination remains effectively unchanged compared to D2, with test AUC values tightly clustered around 0.71.

Table 5.4: Predictive performance across model families for D3 (tuned design).

Model	CV AUC	Train AUC	Val AUC	Test AUC	Brier Score	F1 Optimal
Logistic Regression	0.716	0.718	0.717	0.712	0.212	0.459
XGBoost	0.718	0.736	0.719	0.713	0.209	0.462
Random Forest	0.717	0.766	0.718	0.713	0.205	0.462
MLP	0.713	0.743	0.710	0.712	0.207	0.467

5.2.2 Cross design comparison

The best test AUC increased from 0.582 in D1 (achieved by MLP) to 0.715 in D2 by XGB, representing an improvement of 0.133, demonstrating the benefit of incorporating pooled longitudinal transitions.

Reducing the feature set to the top-20 predictors in D3 maintained this gain, with XGB achieving a test AUC of 0.713, only 0.002 lower than D2, confirming that most of the predictive signal is concentrated in a small subset of variables, and that performance is robust to feature reduction.

Across designs, best CV performance also improves and stabilises:

- D1: 0.589 ± 0.072 (LR)

- D2: 0.716 ± 0.007 (XGB)

- D3: 0.718 ± 0.006 (XGB)

The drop in CV standard deviation from 0.072 (D1) to 0.007 (D2) reflects a structural shift in the learning problem, not just a larger sample. D1 predicts a clinically coded outcome eight years ahead from a single wave snapshot. The predictive signal is sparse, the outcome is heterogeneous, and fold-to-fold variance dominates. D2 pools 14,296 person wave observations across four two year transitions in which self-reported health burden is the direct outcome. Aligning the predictor horizon with the outcome window drives the improvement. For short-term incident prediction in survey cohorts, self-reported health state is a stronger signal source than objective biomarkers measured years earlier.

5.2.3 Model comparison

Once temporal structure is introduced (D2 and D3), model differences narrow. Test AUC ranges shrink from 0.048 (D1) to 0.016 (D2) to 0.001 (D3), with all models clustering in the range 0.69–0.72 and no single model class dominating. XGB achieved the highest test AUC in both D2 (0.715) and D3 (0.713), though the margin over LR was narrow (≤ 0.006). The fact that the linear LR remained consistently competitive (0.709 in D2 and 0.712 in D3) suggests that risk signals in the ageing population are primarily linear. MLP slightly underperforms in D2 (0.699) but matched other models in D3 (0.712), indicating improved performance when the feature space is reduced to the most informative predictors. This indicates that individual burden indicators, particularly medication count and pain, dominate the risk score with largely linear effects, leaving limited room for nonlinear algorithms to outperform a well regularised linear baseline. The convergence of model families in D3, where the AUC range narrows to just 0.006, is a further sign that the available signal has been largely captured by the top-20 features. Improving performance from here would require richer data rather than more complex algorithms.

Figure 5.2 compares generalisation gaps (Train−Test) AUC across all three designs. The average train−test AUC gap (across all four models) drops from 0.202 in D1 to 0.030 in D2 and 0.028 in D3, a reduction of over 85%. This demonstrates that the pooled multi-wave design effectively reduces overfitting. D1 exhibits the largest gaps, consistent with weaker signal and higher variance in estimates. In contrast, D2 and D3 show much smaller gaps, indicating improved stability. Tree-based models (XGB and RF) exhibit the largest gaps in D1, consistent with the small sample size. In D2 and D3, XGB achieves the smallest gaps (0.021 and 0.023 respectively), while RF shows comparatively larger gaps (0.029 and 0.053). The strong performance of XGB in D3 is expected, as the top-20 feature set was selected using XGB gain importance from D2. Nevertheless, XGB also generalises well in D2, suggesting that its boosting mechanism is effective at learning stable patterns from pooled longitudinal data.

In contrast, RF may be more prone to overfitting even with the pooled design, possibly due to its reliance on bagging and random feature selection, which may capture informative predictors less efficiently than boosting.

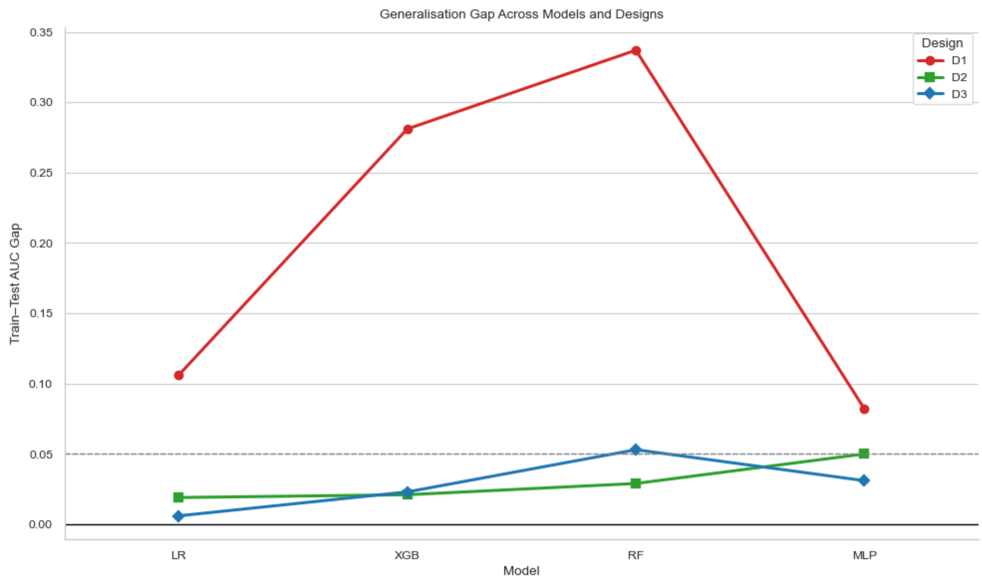

Figure 5.2: Train−Test generalisation gaps across models and designs. The 0.05 dashed line marks the conventional acceptable overfitting threshold.

The figure also highlights that model level differences in generalisation gap are more pronounced in D1, whereas D2 and D3 exhibit convergence across model families, suggesting that data structure plays a larger role than algorithmic choice in controlling overfitting.

5.2.4 Track B: Disease specific

Track B targets individual chronic disease outcomes using the same pooled cohort structure and participant level splitting as D2 and D3, with XGB as the sole classifier given its consistently strong performance in Track A. Twenty-four disease specific outcomes were evaluated using two feature strategies. All eligible predictors retained after the 50% missingness filter (for example, 261 features for functional decline and 351 for circulatory disease), and a restricted top-20 set derived from XGB gain importance. Outcomes span key domains of ageing related health in TILDA, covering cardiovascular conditions, functional decline, mental health, and metabolic disease.

5.2.4.1 Outcome categories

- **Clinical (Incident case diseases (ICD-10)):** Circulatory, endocrine/metabolic, eye, and musculoskeletal diseases (comprehensive from wave 5 only, source predictors excluded to prevent leakage).

- **Functional:** Functional decline, falls (Inc. balance), pain, hearing loss, and urinary incontinence (self-reported, available across multiple waves e.g., W2→W4, W3→W4→W5 and W2→W4).

- **Family history:** Reports of family member diagnoses across eight conditions. These are not incident diagnoses of the participant and represent a distinct prediction target, estimating the likelihood of a participant newly reporting a family history item at follow-up. These outcome variables were retained through the manual filtering process described in Section 3.4.

- **Medication/procedure proxies:** Antidepressant, antihypertensive, cholesterol medications, doctor confirmed hypertension, and cardiac procedure. These are downstream markers of clinical decision making used as prediction targets. They capture risk signals linked to eventual treatment rather than raw disease onset. As noted in Appendix H, incidence rates for medication proxies vary widely, 1.4% for cardiac procedure to 41.6% for doctor confirmed hypertension diagnosis (though this is based on a very small cohort of $n \approx 250$).

5.2.4.2 Overall performance

Figure 5.3 plots (train against test) AUC for all disease specific models, coloured by outcome category. The diagonal line represents perfect generalisation (train equals test), while lines at 0.5 on both axes mark chance level performance. Points close to the diagonal indicate stable learned signal, whereas points far above it, with much higher train than test AUC, indicate overfitting. Overall, test AUC values range from approximately 0.45 to 0.72, reflecting substantial variation driven mainly by outcome type and incidence rate rather than model choice.

Functional and symptom outcomes were the most reliable across both feature strategies. Functional decline ($n = 14{,}296$, incidence 22.5%) achieved a test AUC of 0.713 with a small gap between train and test. This outcome uses a similar self-reported chronic health problem variable as D2 and D3 in Track A derived from the variable codebook (not CAPI), and the consistent performance across both tracks confirms the stability of this predictor outcome relationship and persistent FR signal identified in the literature. FOF (n = 18,114, incidence 14.0%) achieved 0.721, the highest test AUC, closely followed by antihypertensive prescription, reflecting the well documented link between functional limitation and falls risk in TILDA. Hearing loss (n = 10,780, incidence 15.1%) reached 0.678 (in TILDA this has been independently associated with cognitive decline [46], making accurate onset prediction clinically meaningful), and urinary incontinence (n = 16,208, incidence 8.4%) achieved 0.682, both with gaps below 0.10. These outcomes benefit from consistent measurement across waves, sufficient event counts, and clear physiological pathways, which support more stable longitudinal prediction.

Medication proxies achieved competitive test AUC values. Antihypertensive prescription (n = 19,845, incidence 4.5%) reached 0.718, and antidepressant prescription (n = 4,378, incidence 3.2%) reached 0.702, despite low incidence. Both models showed train AUC above 0.99, indicating overfitting to training specific patterns rather than fully generalisable risk patterns. The test results show that the models can discriminate between cases and non cases, although part of this performance likely comes from established treatment profiles. For example, a participant already receiving antihypertensives at baseline has a clinical profile strongly associated with future prescription continuation. These models are therefore informative for risk stratification but should not be interpreted as predicting new disease onset.

Clinical coded ICD-10 outcomes showed moderate performance. Circulatory disease (n = 14,539, incidence 12.2%) and eye diseases (n = 4,396, incidence 10.1%) were the strongest, reaching test AUC values of 0.626 and 0.653, respectively. This aligns with findings that cardiovascular outcomes in TILDA are supported by well established clinical predictors. Musculoskeletal diseases (n = 4,269, incidence 6.5%) and endocrine / metabolic diseases (n = 4,117, incidence 8.8%) performed near chance, with test AUC values of 0.507 and 0.515 under the all features strategy, alongside train AUC values of 1.000 and 0.967. This pattern again indicates that the models memorised training labels without learning signal that transfers to new data. This issue is discussed further in Section 5.2.4.4.

Family history models were the least reliable. Test AUC ranged from 0.513 (cancer n = 3,971, incidence 13.4%) to 0.665 (high cholesterol n = 4,312, incidence 12.0%), while training AUC reached 1.000 for several outcomes such as Alzheimer's disease, indicating severe memorisation. Switching to the top-20 feature set consistently degraded performance, with several models dropping below 0.50. The root cause is structural. Family history features capture a participant's reported knowledge of a relative's diagnosis, not an outcome the participant has experienced. There is no direct causal pathway linking participant-level predictors to these targets, so what the model learns from training does not transfer to unseen data.

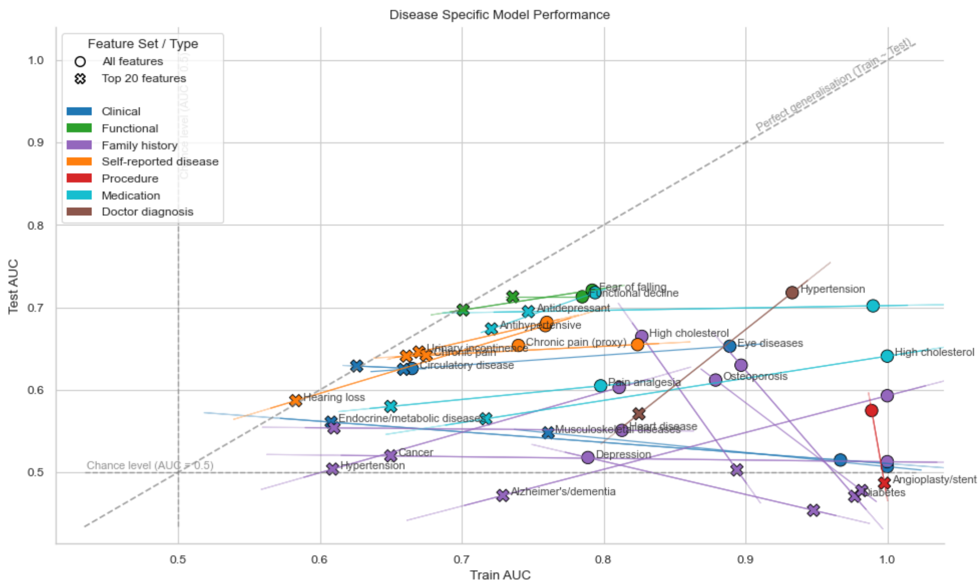

Figure 5.3: Train vs Test AUC for all Track B disease specific models, coloured by outcome category. For each disease, a circle marks the all features model, and a cross marks the top-20 models. The connecting line shows the shift between the two strategies for the same disease. Dashed lines at 0.5 indicate chance level discrimination. Shaded ellipses show the spread of points within each outcome category.

5.2.4.3 Incidence and predictability

Cohort sizes and incidence rates for each outcome are provided in Appendix H. Incidence rate appears to be the most consistent driver of model stability in Track B. Outcomes with incidence above approximately 12%, particularly when measured consistently across waves, tended to achieve test AUC values above 0.65. In contrast, outcomes with incidence below approximately 7% generally produced near chance test AUC and large gaps between train and test regardless of feature strategy.

A model trained on few positive examples cannot learn stable decision boundaries. Class imbalance weighting via `scale_pos_weight` partially compensates, but when positive cases are very sparse, the model can achieve high training scores by memorising those few cases without generalising. This is why Train = 1.000 appears for musculoskeletal diseases, Alzheimer's disease, and cancer in the all features models, none of which exceed test AUC of 0.54 despite high training performance.

Doctor confirmed hypertension is an outlier worth noting. Its incidence rate of 41.6% is the highest in Track B, and it achieved test AUC of 0.718. However, the cohort is very small at $n = 250$, making this estimate less reliable. Given the small cohort, this estimate carries uncertainty and needs careful interpretation, as test scores at this scale are sensitive to which participants happen to fall in the test split.

5.2.4.4 Overfitting and feature reduction

For outcomes where the all features strategy produced train AUC substantially above 0.90 alongside near chance test AUC, the top-20 strategy provides a more reliable estimate of generalisable performance, even when absolute performance remains modest. Musculoskeletal diseases moved from train 1.000 / test 0.507 to train 0.761 / test 0.548 under feature reduction. Endocrine and metabolic diseases similarly improved from 0.515 to 0.561. These are not large gains, but they indicate a more calibrated model that has learned from signal rather than noise. In contrast, for outcomes where both strategies produced similar test AUC with small gaps, such as functional decline, FOF, and hearing loss, the feature reduction confirms stable learning rather than addressing overfitting.

For family history targets, feature reduction worsened rather than helped. With only 20 features, the model is further constrained in its ability to identify even proxy signal for family reported diagnoses. This is consistent with the broader point that the target type, rather than the feature count, drives instability in these models.

The overall pattern reflects the combined effect of two filtering stages. The manual screening in Section 3.4 permanently removed variables exceeding the 80% missingness threshold. This improved overall signal quality but also narrowed the predictor pool, mostly for outcomes that depend on variables which were rare or inconsistent across waves. Within the ML pipeline, a separate 50% missingness filter is applied per outcome on training data only, further constraining available features for outcomes with sparse or indirect measures. Outcomes with rich, well distributed predictor signals, such as functional decline and FOF, benefited most. Outcomes dependent on rare or indirect signals, such as Alzheimer's family history or cardiac procedure, remained constrained by both the target definition and the reduced predictor space. The results in Figure 5.3 should therefore be interpreted not only as a function of modelling choices, but as a reflection of data availability and the structural properties of each outcome category in TILDA.

5.3 Model interpretability and feature importance

Feature importance is reported for all four model families in Appendix J, with predictor add-on results in Figure 5.4. Signal concentrated in a compact cluster of health burden features regardless of design. The dominant predictors shift with the prediction window, and no domain specific add-on contributed statistically significant independent value.

Global SHAP plots for D1, D2, and D3 across all four model families, together with four disease specific XGB models from Track B, are provided in Appendix K. In D1, BP, anthropometric markers, and attendance at community classes or lectures carry the highest mean |SHAP| values, consistent with the 8 year window ICD-10 prediction task. The binary hypertension flag ranks fourth in LR and MLP (|SHAP| ≈ 0.025) but contributes zero under XGB. As a tree model, XGB captures the same cardiovascular information directly from continuous BP readings rather than from a derived indicator. Attendance at community classes or lectures appears in the D1 top 5 across all models but drops out entirely in D2. This pattern aligns with Section 2.3, community participation carries predictive signal over an 8 year window but not over 2 year transitions in self-reported health. In D2, medication count produces the strongest shift under XGB (|SHAP| = 0.194), while self-rated health leads in LR (0.118), showing that linear and tree-based models weight the same predictors differently. University of California Los Angeles (UCLA) loneliness appears at rank 16 in the LR (|SHAP| = 0.034) but is absent from the tree and neural models, consistent with the near zero add-on delta for loneliness reported in Section 5.3.4. A weak linear association is absorbed by correlated pain and healthcare utilisation predictors once ensemble models are used.

The Track B SHAP plots (also in Appendix K.3) show predictor structures that diverge considerably from the Track A pattern, reflecting the measurement gap between self-reported and clinically coded diseases. Functional decline mirrors the Track A burden hierarchy (self-rated health 0.172, medication count 0.166, chronic pain proxy 0.137). FOF is instead driven by age (0.232), physical impairment count (0.207), and two anxiety measures (0.110, 0.071), none of which appear among the top-20 gain features from D2, consistent with the cognitive and emotional dimension of FOF reviewed in Chapter 2 [22]. For hearing loss, the pre-existing hearing difficulty item (|SHAP| = 0.367) was more than twice as influential as the second ranked predictor (0.139), confirming the model captures progression of an existing condition rather than new onset. Circulatory disease is age dominated (0.150), with retirement status (0.031) in the top five. Each outcome draws on different signal sources, which explains why no single compact predictor set generalises across all Track B disease targets.

5.3.1 Clinical biomarkers dominate

In D1, the top-20 features selected by `SelectKBest` were dominated by cardiovascular and anthropometric measurements from the wave 1 health centre assessment (see Appendix J.1). BP variables, covering seated and standing systolic and diastolic readings and their means, appeared prominently across all four model families. Anthropometric FR markers, specifically hip circumference (0.084 for RF, 0.037 for MLP), waist circumference, and BMI, were consistently ranked alongside trail making test time as a marker of cognitive processing speed. Social engagement through attendance at classes or lectures, health insurance coverage, mother's age at death as a familial longevity signal, socioeconomic group, and physical impairment count contributed additional signal across models.

BP is a well established cardiovascular risk marker [35], anthropometric measures reflect the FR and obesity pathway identified in TILDA [21], and trail making test time captures cognitive processing speed [27]. These objectively measured features dominated D1 because wave 1 included a full health centre assessment, providing clinical measurements unavailable or inconsistently recorded in later waves.

Across model families, the ranking order varied modestly. Seated systolic BP ranked first for both XGB (gain 0.072) and RF, consistent with its known role as a cardiovascular risk marker [35] and its consistent identification across tree-based approaches. By contrast, LR placed attendance at community classes or lectures at rank one (0.039), and MLP placed hip circumference first (0.037). The prominence of community class attendance in LR is consistent with the literature on social engagement as a long horizon health indicator. Socially active adults at baseline are less likely to develop chronic disease over an 8 year window [32]. The cross model consistency in which features appear, rather than their specific scores, is the more meaningful observation.

5.3.2 Medication burden and self-rated health emerge

Shifting to D2 produced a major change in the importance landscape (Appendix J.2). BP variables, which dominated D1, do not appear in the top-20 for any model in D2. Medication count excluding supplements ranks first in three of four model families (importance score 0.068 for XGB), with polypharmacy indicators occupying much of the top 10. Self-rated general health ranked first for LR (importance score 0.150) and appeared in the top six across all model families. Comparative self-rated health, capturing how participants assess their health relative to same age peers, also featured across LR, XGB, and RF. Pain variables, including limiting pain, severity, and any chronic pain indicator, contributed consistently across models, as did functional disability items such as chair rising difficulty, mobility limitation counts, and walking or standing steadiness. Healthcare utilisation (General Practitioner (GP) and outpatient visits) rounds out the top-20, reflecting that participants already engaging heavily with the health system carry elevated near term risk. The shift is explained by the prediction window. Over 8 years, clinical measurements reflect long-term biological risk. Over 2 years, medication burden (a participant already taking multiple medications at baseline is substantially more likely to report chronic illness at the next wave) and self-rated health are more direct markers of near term change.

5.3.3 Cross model agreement and feature set

The four model families agree substantially on the top predictors in D2. The same health burden features appear across LR, XGB, RF, and MLP with only minor ordering differences. This convergence across methods strengthens confidence in the identified cluster. Table 5.5 lists the D2 XGB gain features applied in D3 and Track B.

The feature set spans two primary thematic domains. Physical/Clinical (medication counts, self-rated health, healthcare utilisation) and Functional (mobility difficulty, pain, physical limitation), with one Lifestyle item (flu vaccination). Nutritional and metabolic biomarkers, psychological and cognitive measures, and sociodemographic features are absent from the top-20. Cholesterol, the nutritional biomarker that did enter the add-on analysis, showed no statistically significant independent contribution (ΔAUC: $[-0.011, 0.023]$), consistent with the broader null finding across domains.

5.3.4 Predictor add-on analysis

The add-on analysis evaluated the incremental contribution of individual candidate features to the prediction of functional decline using a *BASE* versus *BASE+*

Table 5.5: Top-20 features from XGBoost gain in D2, applied in D3 and Track B.

Feature label	Description	Thematic domain
Reported medications (excl. supps)	Total reported medications excluding supplements	Physical/Clinical
Chronic pain (proxy)	Frequency of pain experienced	Functional/Symptom
Polypharmacy (excl. supps)	Taking five or more medications excluding supplements	Physical/Clinical
Regular medications	Number of regular medications including supplements	Physical/Clinical
Reported medications	Number of reported medications including supplements	Physical/Clinical
Self-rated health	General self-rated health (excellent to poor)	Physical/Clinical
Number of physical limitations	Count of activity limitations from mobility items	Functional
Health limits activities	Health problem limits kind or amount of activities	Functional
Pain limits activities	Chronic pain restricts daily activities	Functional/Symptom
GP visits (12 months)	Number of GP visits in past 12 months	Physical/Clinical
Large muscle ADL/IADL index	Composite of large muscle daily activity items	Functional
Pain severity	Level of pain reported	Functional/Symptom
Flu vaccination	Had a flu vaccination in past 12 months	Lifestyle
Walking/standing steadiness	Steadiness when walking or standing	Functional
Chronic pain	Any chronic pain present	Functional/Symptom
Chair rising steadiness	Steadiness when getting up from a chair	Functional
Hospital visits (12 months)	Number of outpatient visits in past 12 months	Physical/Clinical
Getting up from chair difficulty	Difficulty getting up from a chair after sitting	Functional
Functional difficulty	Difficulty stooping, kneeling, or crouching	Functional
Health vs peers	Self-rated health compared with others of same age	Physical/Clinical

'Chronic pain (proxy)' includes proxy respondents and is derived from CAPI. 'Chronic pain' is a separate binary indicator from the derived variable codebook.

ADDON framework. For each candidate, a model was trained on all available predictors except that candidate (BASE), then retrained with it included. The baseline cohort and feature set are determined dynamically for each candidate. Only waves in which both the outcome target and the candidate feature were present were used, meaning cohort size and feature count varied across candidates. For example, when a history of blackout or fainting served as the candidate feature, the analysis included waves 1 to 3, yielding 929 held-out test participants and approximately 570 features entering the model after the missingness filter. When education level served as the candidate, waves 3 to 5 were used, yielding 660 test participants and approximately 464 features. The delta in AUC and (AP) was estimated with bootstrapped 95% CIs (B = 2,000 resamples on the held out test set). Full delta results are shown in Figure 5.4. The candidates span lifestyle, socioeconomic, cognitive, nutritional, and social domains, including features from thematic domains listed in Table 3.2 that are not present in the top-20 feature set.

This analysis tests whether any domain contributes independent predictive information beyond the established baseline. Given the large baseline feature sets, marginal gains from a single candidate are expected to be small. No candidate produced a statistically significant improvement across any metric. Formally, for each candidate the null hypothesis $H_0\colon \Delta\text{AUC} = 0$ was tested, rejection requires the interval to exclude zero. Every confidence interval contained zero, meaning no candidate produced a statistically significant gain. Selected results, organised by domain cluster, are as follows.

(i) **Functional predictors** (chair rising difficulty, physical limitations, chair rising steadiness, FOF). Among the functional candidates, the highest BASE AUC occurred when getting up from a chair difficulty served as the candidate feature (0.707). Adding it to the BASE produced a near zero marginal gain (CI: $[-0.012, 0.002]$). Physical limitation count (CI: $[-0.008, 0.007]$) and chair rising steadiness (CI: $[-0.006, 0.010]$) showed similarly null deltas. As reviewed in Table 2.2, UGS, chair stand performance, and physical limitation are consistently identified as primary FR indicators in TILDA [7]. The small incremental contributions are consistent with strong correlation across functional predictors: pain, steadiness, and mobility difficulty items share their predictive variance, and none appears to contribute substantial independent information beyond the others.

(ii) **Socioeconomic predictors** (education level, household income). Education level produced the largest observed positive delta, with Test AUC moving from 0.711 to 0.719 (CI: $[-0.002, 0.018]$). The direction is consistent with the examined literature on social disadvantage [15], although the interval crosses zero and the effect cannot be confirmed statistically. Total household income produced a smaller positive trend ($[-0.004, 0.010]$). The smaller effect likely reflects alignment with healthcare utilisation and employment status predictors already in the BASE.

(iii) **Psychosocial and mental health predictors** (UCLA loneliness, depressive symptoms, social connectedness). UCLA loneliness produced no detectable effect (CI: $[-0.009, 0.006]$). Depressive symptoms produced a small positive delta of $+0.001$ (CI: $[-0.008, 0.010]$). Social connectedness showed similarly minimal movement (CI: $[-0.008, 0.007]$). Section 2.3 documents bidirectional links between SOC (social isolation, depression, and functional health) in TILDA [22, 32]. The null deltas do not contradict these associations, they indicate that the relevant pathways are already captured within the functional and pain predictors present in the BASE.

(iv) **Physical and behavioural predictors** (GRIP, BMI, cigarettes per day, CAGE alcohol scale, history of blackout or fainting). HGS produced a positive delta of +0.004 (CI: $[-0.004, 0.012]$). Chapter 2 identifies GRIP as a primary FR marker in TILDA [18]. The near zero increment confirms that the functional limitation variables in the BASE already capture this FR related signal. BMI produced +0.002 (CI: $[-0.007, 0.012]$). Cigarettes per day trended slightly negative ($[-0.012, 0.002]$), and the CAGE alcohol scale contributed minimally ($[-0.004, 0.008]$). A history of blackout or fainting produced the highest BASE AUC in the study (0.714, CV 0.726 ± 0.009), indicating a BASE model that is already performing strongly when this variable is withheld. Adding it back yielded no further gain (CI: $[-0.007, 0.007]$), suggesting its signal is captured through correlated cardiovascular and health burden predictors already present in the BASE.

Predictive signal in this cohort is concentrated in a compact, internally correlated set of health burden indicators. Restricting D3 to 20 features caused no performance loss, and no add-on candidate improved discrimination significantly. Domain specific features, whether socioeconomic, cognitive, nutritional, or behavioural, add no independent value beyond the core burden set for two year transitions in this population.

5.4 Health burden correlations

The performance patterns in this chapter are partly explained by the structure of the underlying data. Pairwise correlations among key health burden features explain three recurring findings: why pooled designs outperformed the single wave baseline, why feature reduction caused negligible performance loss, and why no add-on candidate improved discrimination significantly.

When pairwise correlations among predictors approach $|r| = 1$, the effective rank of the feature matrix decreases and the marginal information gained from

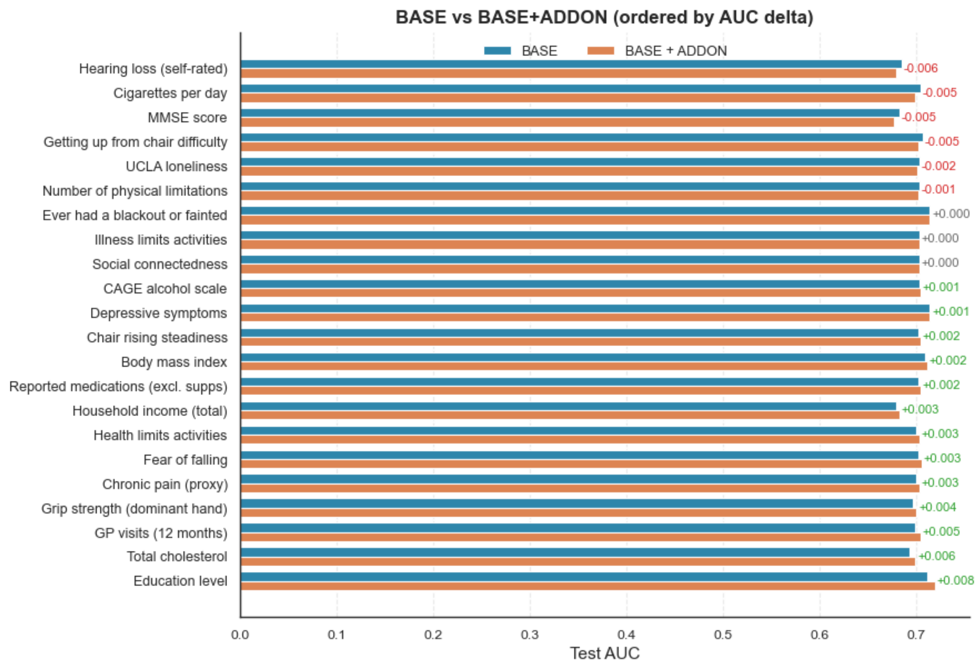

Figure 5.4: Bootstrapped 95% confidence intervals for test AUC delta (BASE+ADDON minus BASE) across all candidate features (B = 2,000 resamples). Features are sorted in ascending order by the midpoint of the delta interval.

adding further correlated features approaches zero, a property that directly explains why the feature reduction in D3 produced negligible performance loss, and why no add-on candidate in Section 5.3.4 improved discrimination significantly. Figure 5.5 presents pairwise Spearman correlations at wave 5 (n = 4,978) for six health burden features used across Track A and Track B: self-reported long term health problem, long term illness, any clinically coded incident cases of chronic disease, chronic pain, self-rated health, and medication use.

The most striking feature of the matrix is the perfect correlation (r = 1.00) between self-reported long term health problem and long term illness. Both share identical prevalence (44.2%, n = 2,201 out of 4,977 valid responses) and are effectively the same variable within this wave of TILDA. They encode a single construct, and predictive performance for either target reflects the same signal rather than two independent outcomes.

A more consequential pattern concerns the alignment between the composite

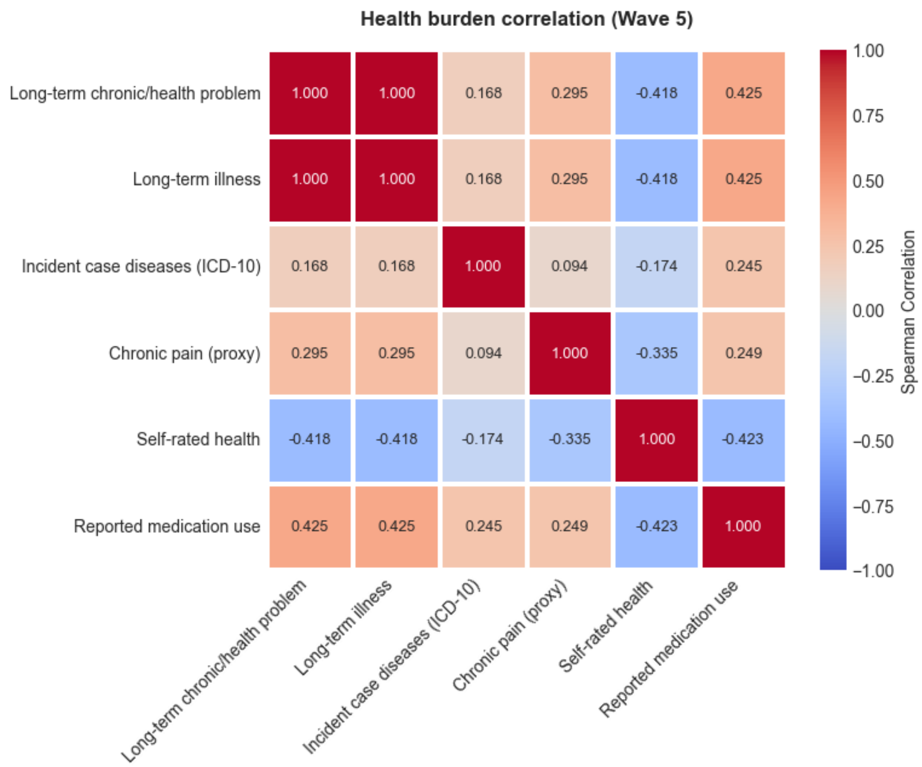

Figure 5.5: Pairwise Spearman correlations among six health burden features at wave 5 (n = 4,978). Self-reported long term illness and chronic health problem form an identical construct (r = 1.00), while any ICD10 shows consistently weak associations with all self-reported measures.

ICD-10 disease indicator and the self-reported measures. Correlations range from $r = 0.094$ with chronic pain to $r = 0.245$ with medication use, with long term illness at $r = 0.168$ and self-rated health at $r = -0.174$. These weak associations are not driven by imbalance, the ICD-10 composite indicator prevalence was 37.6%, comparable to the self-reported long-term health problem prevalence of 44.2% reported above. They reflect a genuine measurement gap, and as confirmed by the Cohen's Kappa analysis in Section 5.1.1 ($\kappa = 0.167$), the agreement between them is poor.

All top-20 features identified in D2 and D3, including medication burden, self-rated health, chronic pain, mobility limitation, and healthcare utilisation, are survey derived and naturally aligned with the self-reported layer rather than with clinically coded records. The issue is not that these predictors are uninformative, but that they are aligned more closely with subjective and functional health states than with clinically coded disease definitions. This additionally explains why D1, which used ICD-10 composite outcome, produced substantially weaker and less stable discrimination than D2 and D3.

A third pattern concerns the internal structure of the self-reported measures. Medication use, self-rated health, chronic pain, and long term illness are moderately correlated with pairwise values ranging from $|r| = 0.25$ to $|r| = 0.43$. Self-rated health is inversely associated with all other burden indicators, medication use ($r = -0.423$), chronic pain ($r = -0.335$), and long term illness ($r = -0.418$). These moderate correlations indicate a shared burden signal, while still leaving each measure with some independent predictive value.

This internal structure directly explains why the add-on analysis produced no statistically significant results. Add-on candidates from outside this set, including education level, GRIP, UCLA loneliness, and cognitive function (MMSE, the survey based cognitive screener available across all waves), were each tested against a BASE model already containing the core burden indicators. None produced a measurable increment, because the principal signal had already been absorbed by these features.

Prediction of chronic disease risk in this cohort is therefore governed by a compact and internally correlated set of perceived and functional health measures. Further gains from adding variables of the same type are unlikely. Richer data that could distinguish perceived health burden from objective physiological decline, including linked mortality records, primary care records, and repeated biomarker panels, are held under elevated access and were not available for this study. The inclusion of family history indicators in Track B partly reflects this constraint, participant level onset data for specific conditions were often sparse or absent in the public release, and family reported diagnoses served as an indirect proxy where direct measures were unavailable.

In this cohort, predicting chronic disease onset is largely a function of existing health burden. Medications, pain, and self-rated health carry the dominant signal, not specific biomarker or lifestyle profiles. Extending beyond this requires richer linked data, as outlined in Chapter 6.

Chapter 6

Conclusion

6.1 Summary

This study developed and evaluated a longitudinal ML pipeline for predicting chronic disease onset in older adults across all five waves of TILDA data. Manual variable screening reduced over 5,000 raw variables to a structured, domain organised dataset. A deterministic cleaning pipeline was applied and validated against published benchmarks. Four model families were evaluated across three modelling designs and 24 disease specific outcomes. Pooled multi-wave designs substantially outperformed the single wave baseline, predictive signal concentrated in a compact cluster of health burden indicators, and no domain specific add-on feature contributed statistically significant independent predictive value. Code and pipeline scripts are available via the repository referenced in Section 1.3.

6.2 Conclusions

Pooled longitudinal designs outperform the single wave baseline, and expanding beyond the core health burden predictor set adds no statistically significant gain in performance. The gains documented in Chapter 5 reflect a structural alignment between predictor horizon and outcome definition. Predicting clinically coded disease eight years ahead is a harder task than predicting self-reported health burden over two-year transitions, and matching these two layers is what drives the improvement. Chapter 2 framed prediction as estimating $P(\text{disease} \mid X)$, where the marginal gain from any added feature depends on how much independent information it carries. The domain counts confirm this. Of the 3,113 retained variables (Table 3.4), sociodemographic context (844), psychological and cognitive measures (291), and nutritional biomarkers (22) produced no top-20 predictors. Lifestyle behaviours (120 variables) contributed one item, flu vaccination, which functioned here as a proxy for healthcare engagement rather than a modifiable behaviour. Signal was concentrated in physical health (853 variables) and functional status (174 variables), and just 20 of those 1,027 sufficed. Adding 1,277 variables across the remaining four domains, each genuinely relevant to ageing, did not move the posterior.

The validation in Chapter 4 adds a dimension often missing from published work. Encoding decisions can materially affect model performance in health survey data. Recoding six frailty index items to respect published ordinal deficit scoring improved AUC from 0.591 to 0.636 under the biologically informed proxy, a 7.6% gain attributable entirely to encoding with no change in the feature set or model.

The three design structure provides a direct test of what drives the performance improvement. Adding temporal structure (D1 to D2) improved performance substantially. Restricting features to 20 (D2 to D3) produced no further loss, confirming that the signal is compact and that the cross-design gap reflects longitudinal structure, not feature count. Hyperparameter optimisation in D3 similarly produced negligible gains.

Track B reinforces the pattern. Outcomes with sufficient incidence and consistent wave measurement generalised well, those depending on sparse events or indirect proxy targets did not, regardless of algorithm. Prediction is bounded by the depth of the data, not the complexity of the model. Significant gains require the linked records held under elevated access, as outlined in Section 6.6.

6.3 Data availability

This analysis was conducted using the publicly accessible TILDA release (waves 1–5), a pseudonymised dataset covering self-reported health, physical, cognitive, nutritional, and behavioural variables, together with clinical coded disease indicators from wave 2 onwards. The datasets are accessible through ISSDA at `https://tilda.tcd.ie/data/accessing-data/`.

Several data sources relevant to chronic disease prediction are not included in the public release. These require on-site access at TILDA's Trinity College Dublin facility or remote access through the TILDA VISTA trusted research environment, and include a dedicated biomarkers dataset, the wave 3 MRI dataset, an epigenetics dataset, linked primary care records, and linked mortality records. Access is subject to formal application to the TILDA data access committee and operates on a cost recovery basis where institutional funding is available. Future work drawing on these sources would require updated ethics approval, GDPR training certification and a signed data use agreement with TILDA.

6.4 Ethics statement

The TILDA study was reviewed and approved by the Faculty of Health Sciences Research Ethics Committee at Trinity College Dublin, and all participants provided written informed consent to their data being used for research purposes. The secondary analysis conducted in this research was approved by the University of Limerick Science and Engineering Research Ethics Committee. All data used were publicly released, pseudonymised datasets accessed through ISSDA. No direct participant contact or identifiable personal data were involved in this analysis.

6.5 Contributions

This research makes contributions at three levels. **Empirically**, it demonstrates that pooled longitudinal modelling substantially improves predictive performance and stability for chronic disease onset in the TILDA cohort, and identifies a compact cluster of self-reported health burden indicators as the primary driver of that signal. Domain features outside this core set showed no statistically significant independent predictive value, a result consistent across all model families and designs. **Methodologically**, a transparent, reproducible preprocessing pipeline for aligning heterogeneous longitudinal health survey data is documented in full, addressing a gap rarely covered in published ML literature. **Clinically**, the results provide a data driven basis for risk stratification using publicly available survey data, and identify the conditions under which more complex predictor sets or model families are unlikely to improve performance without access to richer linked data.

The results in Table 2.2 provide a concrete benchmark. Test AUC values in D2 and D3 (0.71–0.72) sit at the upper end of the 0.64–0.71 range reported in comparable longitudinal prediction studies, including AUC $\leq$ 0.69 for falls prediction in TILDA [19] and 0.64–0.71 for longitudinal mental health disorder prediction [26]. The present results are comparable with, and in some cases above these benchmarks while targeting a broader outcome using only publicly accessible data. The single wave baseline (D1: 0.53–0.58) confirms that the gain is driven by the pooled longitudinal design, not algorithm choice. All four model families cluster within 0.001 AUC in D3, which reinforces this.

The reported benchmark of 0.854 [25] reflects both an unclear data context and a different modelling architecture. Predicting 2 year diabetes onset using 105 curated predictors selected from over 40,000 clinical parameters, combined into a stacked ensemble of five algorithm families including DNN. This study evaluated disease specific outcomes in Track B, including diabetes related conditions, using only the variables available in the public TILDA release and individual model families rather than a stacked ensemble. The performance gap reflects both the depth of the available predictor set and the architectural differences in the modelling approach.

6.6 Future work

The recently published sixth wave [47] of TILDA data (December 2025), together with the Intellectual Disability Supplement (IDS) and the COVID-19 [48] sub study, would extend the longitudinal coverage and introduce health trajectory data spanning a pandemic period, likely altering self-reported burden profiles in meaningful ways. Cross cohort validation using the RAND harmonised TILDA dataset, which maps variables to international standards compatible with cohorts such as HRS and SHARE, would test how well the identified health burden cluster generalises beyond the Irish context. Subgroup analyses would also strengthen the clinical utility of these findings. Testing groups based on initial FR status will show if the predictors hold true for all levels of FR, informing which individuals are most likely to benefit from early intervention. Stratifying by polypharmacy burden at baseline would clarify whether the model retains predictive power in participants not yet medicated, which is where prevention strategies carry the greatest potential impact. Analyses by age group and sex would further characterise whether the relative importance of functional and clinical indicators shifts across these subgroups, given known sex differences in chronic disease incidence within TILDA. Finally, access to the restricted biomarker panels, primary care records, and linked mortality data held within TILDA's secure research environment would allow future work to test whether the signal limits identified here reflect the nature of chronic disease risk itself or the constraints of the publicly available dataset.

The encoding challenges encountered in preparing the TILDA datasets point to a wider gap in the ML literature. Standardised preprocessing protocols and auditable encoding indices for longitudinal health survey data do not yet exist as a shared resource, and creating one would face real obstacles. Health survey variables differ substantially in format, coding convention, and clinical meaning across cohorts, and privacy regulations limit direct data sharing across institutions. A practical contribution would be a curated methodological guide documenting principled approaches to encoding heterogeneous health data for ML, covering transformation rationale, audit trail requirements, and domain specific considerations. Such a resource would reduce the risk that observed performance differences reflect encoding choices rather than genuine signal, and provide clearer guidance for researchers working with comparable survey based cohorts.

Appendix A

Script for loading/exporting PMF metadata to CSV

```python
import pandas as pd
import pyreadstat
from pathlib import Path
import json

BASE_DIR = Path(__file__).resolve().parent
# =TEXTJOIN(", ", TRUE, J2:Jxxx) - Excel formula for number of
    xxx rows from manual filtering to create var of interest with
    comma
data_files = {
    "wave1.json": BASE_DIR / "data/0053-01_TILDA_Wave1_2009
        -2011/0053-01_TILDA_Wave1_Data/SPSS/0053-01
        _TILDA_Wave1_PMF_v1.12.sav",
    "wave2.json": BASE_DIR / "data/0053-01_TILDA_Wave2_2012-2013/
        Data/0053-02_TILDA_Wave2_PMF_v2.7.sav",
    "wave3.json": BASE_DIR / "data/0053-01_TILDA_Wave3_2014-2015/
        Data/0053-04_TILDA_Wave3_PMF_v3.6.sav",
    "wave4.json": BASE_DIR / "data/0053-01_TILDA_Wave4_2016/Data
        /0053-05_TILDA_Wave4_PMF_v4.4.sav",
    "wave5.json": BASE_DIR / "data/0053-01_TILDA_Wave5_2018/Data
        /0053-06_TILDA_Wave5_PMF_v5.5.sav",
    "harmonized.json": BASE_DIR / "data/0053-06
        _TILDA_Harmonized_2016/Data/0053-03_TILDA_Harmonized_v1.2.
        dta",
```

```python
}

def load_df_with_var_of_interest(var_dir="var_of_interest", data_files=data_files, verbose=False):
    dfs = {}
    var_dir = BASE_DIR / var_dir

    for json_name, data_path in data_files.items():
        json_file = var_dir / json_name

        with open(json_file, "r") as f:
            vars_list = json.load(f).get("variables_of_interest", [])

        vars_list = [str(v) for v in vars_list]
        if verbose:
            print(f"Variables of interest from {json_file.name}: {vars_list[:10]} ...")

        # ---------- SPSS ----------
        if data_path.suffix == ".sav":
            # metadata only (fast, avoids encoding errors)
            _, meta = pyreadstat.read_sav(data_path, metadataonly=True, encoding="latin1")
            original_rows = meta.number_rows
            original_cols = len(meta.column_names)

            # load only needed variables (fast)
            df, _ = pyreadstat.read_sav(
                data_path,
                usecols=vars_list,
                encoding="latin1",
                apply_value_formats=True,
            )

        # ---------- STATA ----------
        elif data_path.suffix == ".dta":
            df_full = pd.read_stata(data_path, convert_categoricals=True)
            original_rows, original_cols = df_full.shape
```

```
            existing_vars = [v for v in vars_list if v in df_full.columns]
            df = df_full[existing_vars]

        else:
            raise ValueError(f"Unsupported file type: {data_path.suffix}")

        # --- filter variables ---
        existing_vars = df.columns.tolist()
        missing_vars = sorted(set(vars_list) - set(existing_vars))
        if missing_vars:
            print(f"Missing variables in {data_path.name}: {missing_vars}")

        # ---------- Verbose print ----------
        if verbose:
            print(f"Data shape (Original): {original_rows} rows x {original_cols} columns")
            print(f"Data shape (Cleaned): {df.shape[0]} rows x {df.shape[1]} columns")
            print("------------------------------------------------")

        dfs[json_name.replace(".json", "")] = df

    return dfs

if __name__ == "__main__":

    output_dir = BASE_DIR / "output"
    output_dir.mkdir(exist_ok=True)

    for json_name, path in data_files.items():
        wave_name = json_name.replace(".json", "")

        if path.suffix == ".sav":
            df, meta = pyreadstat.read_sav(path, encoding="latin1", apply_value_formats=True)
        else:
```

```
            df, meta = pyreadstat.read_dta(path, encoding="latin1")

        pd.DataFrame({
            "variable": meta.column_names,
            "label": meta.column_labels
        }).to_csv(output_dir / f"{wave_name}_variable_descriptions.csv", index=False)

        print(f"--------------- wave {path} ---------------")
        print(df.shape)
        print(df.columns[:20])
        print("Variable descriptions, csv file saved successfully -> dir=output/")

        df.to_csv(output_dir / f"{wave_name}.csv", index=False)

    dfs = load_df_with_var_of_interest(verbose=True)
```

Appendix B

Cleaning pipeline scenario tables

Table B.1: Initial raw feature counts categorised by nine scenarios (pre cleaning)

Scenario	**Wave 1**	**Wave 2**	**Wave 3**	**Wave 4**	**Wave 5**	**Harmonised**
Empty / blank values	9	0	7	0	9	0
Numeric with sentinel	49	25	18	24	17	1
Clean numeric	14	27	18	22	34	8
Censored / bracketed numeric	84	71	91	85	85	73
Binary categorical	199	134	162	187	188	17
Multi class categorical / mixed	319	191	223	244	230	214
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	135	34	120	30	28	140
Total Columns	**809**	**482**	**639**	**592**	**591**	**453**

Table B.2: Scenario count progression: post step 1 (empty / blank handling)

Scenario	**Wave 1**	**Wave 2**	**Wave 3**	**Wave 4**	**Wave 5**	**Harmonised**
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	78	31	37	26	20	15
Clean numeric	124	55	123	50	68	134
Censored / bracketed numeric	84	71	91	85	85	73
Binary categorical	202	134	164	187	188	17
Multi class categorical / mixed	321	191	224	244	230	214
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**591**	**453**

Table B.3: Scenario count progression: post step 2 (sentinel standardisation)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	202	86	160	76	88	149
Censored / bracketed numeric	84	71	91	85	85	73
Binary categorical	202	134	164	187	188	17
Multi class categorical / mixed	321	191	224	244	230	214
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**591**	**453**

Table B.4: Scenario count progression: post step 3 (numeric cleaning)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	249	109	204	112	125	149
Censored / bracketed numeric	16	23	17	16	17	73
Binary categorical	202	135	165	188	188	17
Multi class categorical / mixed	341	215	253	276	261	214
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	1	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**591**	**453**

Table B.5: Scenario count progression: post step 4 (censored / ordinal mapping)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	265	132	221	128	142	222
Censored / bracketed numeric	0	0	0	0	0	0
Binary categorical	202	135	165	188	188	17
Multi class categorical / mixed	341	215	253	276	261	214
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	1	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**591**	**453**

Table B.6: Scenario count progression: post step 5 (binary classification)

Scenario	**Wave 1**	**Wave 2**	**Wave 3**	**Wave 4**	**Wave 5**	**Harmonised**
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	467	261	375	305	286	235
Censored / bracketed numeric	0	0	0	0	0	0
Binary categorical	0	6	11	11	44	0
Multi class categorical / mixed	341	215	353	276	261	218
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	1	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**591**	**453**

Table B.7: Final cleaned dataset: post step 6 (multi-class / mixed parsing)

Scenario	**Wave 1**	**Wave 2**	**Wave 3**	**Wave 4**	**Wave 5**	**Harmonised**
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	809	482	639	592	591	453
Censored / bracketed numeric	0	0	0	0	0	0
Binary categorical	0	0	0	0	0	0
Multi class categorical / mixed	0	0	0	0	0	0
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**591**	**453**

Appendix C

Variable scenario detection function

```python
# Sentinel values (numeric and string)
# -1 and -1.0 is sentinel but will be used in the clean pipeline
standardised_sentinels = {'-1', '-1.0', -1, -1.0} #  set -> duplicates removed -> counts correct -> single value detection is zero
numeric_sentinels = [-1, -1.0, -98, -98.0, -99, -99.0,
                     95, 95.0, 96, 96.0, 98, 98.0, 99, 99.0]
string_sentinels = [str(x) for x in numeric_sentinels] + [f"_{x}" for x in numeric_sentinels] + \
    ['na', 'n/a', 'refused', 'missing', 'dk', 'rf', 'refused', "refuse", 'no response', 'nan', 'not answered', 'no answer',
     'not applicable','no consent','dont know',"don't know","didn't answer",'prefer not to say', 'task not done', 'insufficient sample', 'unable to record',
     '__', '*']
blanks = ['', ' ', '\t', '\n']

SCENARIOS = [
    "Empty / blank values",
    "Numeric with sentinel",
    "Clean numeric",
    "Censored / bracketed numeric",
    "Binary categorical",
    "Multi class categorical / mixed",
    "Single value categorical",
    "Special symbol / mixed",
```

```
    "Unknown / complex mixed"
]


# flatten all dtypes across the waves to float and category
def flatten_column_types(df):
    df_flat = df.copy()

    for col in df_flat.columns:
        if pd.api.types.is_numeric_dtype(df_flat[col]): # Numeric columns
            df_flat[col] = df_flat[col].astype(float)  # Convert to float
        else:
            df_flat[col] = (  # Convert to string, strip whitespace, lowercase, then category
                df_flat[col]
                .astype(str)
                .str.strip()
                .str.lower()
                .astype('category')
            )

    return df_flat

def detect_column_scenario_mapping(df, sample_size=10):
    summary = []

    # Regex patterns
    special_symbols_pattern = r'[^A-Za-z0-9\s\.,\-+&/()]' # special symbols excluding common numeric formatting
    bracketed_numeric_pattern = r'\d{1,3}(?:,\d{3})*[--/]\d{1,3}(?:,\d{3})*|\d+\s*[<>+-]' # patterns like "10-20", "30/40", ">=50"
    sentinel_set = set(x.lower() for x in string_sentinels) # Lowercase string sentinels for comparison

    for col in df.columns:
        col_data = df[col].copy()
        dtype = str(col_data.dtype) # get data type as string

```

```
        # Convert to string for sentinel / blank detection
        str_vals = col_data.astype(str).str.lower().str.strip() # lowercase and strip whitespace
        clean_vals = col_data[~str_vals.isin(sentinel_set) & ~(str_vals.isin(blanks))] # remove sentinels and blanks

        # Remove sentinels and blanks for scenario detection only
        vals_no_sentinel = clean_vals[~clean_vals.astype(str).str.lower().isin(sentinel_set)
                                      & ~clean_vals.astype(str).str.lower().isin(blanks)].unique()
        num_unique_no_sentinel = len(vals_no_sentinel)

        # Check if any sentinel values are present
        sentinel_present = str_vals.isin(sentinel_set).any()

        # Detection flags
        empty_detected = str_vals.isin(blanks).any() or len(col_data) == 0

        # Numeric detection: coerce to numeric, NaNs for non-numeric
        numeric_vals = pd.to_numeric(clean_vals, errors='coerce') # coerce non-numeric to NaN
        numeric_like = numeric_vals.notna().all() and len(clean_vals) > 0 # all values are numeric-like after cleaning

        sentinel_detected = str_vals[~str_vals.isin(standardised_sentinels)].isin(sentinel_set).any() # or col_data.eq(-1.0).any() # check for sentinel presence
        censored_detected = clean_vals.astype(str).str.contains(bracketed_numeric_pattern).any() # check for bracketed numeric patterns

        # Binary categorical detection (exactly 2 unique values after cleaning)
        binary_categorical_detected = False
        if dtype == 'category':
            if num_unique_no_sentinel == 2:
```

```
                binary_categorical_detected = True
            elif num_unique_no_sentinel == 1 and sentinel_present:
                binary_categorical_detected = True

        # Multi-class categorical (3+ unique values, object/category)
        multi_class_categorical_detected = num_unique_no_sentinel > 2 and dtype == 'category'

        # Single value categorical
        single_value_detected = num_unique_no_sentinel == 1 and not binary_categorical_detected

        # Special symbols detection (ignore numeric formatting)
        symbol_detected = clean_vals.astype(str).str.contains(special_symbols_pattern).any()

        # Mixed types detection
        mixed_detected = len(set(type(v) for v in col_data)) > 1  # more than one data type present

      # Determine scenario in order of pipeline
        if empty_detected:
            scenario = SCENARIOS[0]
            handling = "Replace numeric/categorical empty/blank with -1.0"
        elif numeric_like and sentinel_detected:
            scenario = SCENARIOS[1]
            handling = "Replace numeric sentinel values with -1.0"
        elif numeric_like and not sentinel_detected:
            scenario = SCENARIOS[2]
            handling = "Keep as numeric"
        elif censored_detected:
            scenario = SCENARIOS[3]
            handling = "Convert numeric ranges to midpoints or keep as categorical"
        elif binary_categorical_detected:
            scenario = SCENARIOS[4]
            handling = "Encode as 0/1 or keep as category"
```

```python
        elif multi_class_categorical_detected:
            scenario = SCENARIOS[5]
            handling = "One-hot encode, label encode, or keep as category"
        elif single_value_detected:
            scenario = SCENARIOS[6]
            handling = "Keep as category / constant column"
        elif symbol_detected:
            scenario = SCENARIOS[7]
            handling = "Inspect and preprocess based on context"
        else:
            scenario = SCENARIOS[8]
            handling = "Inspect manually / complex mixed types"

        sample_values = list(col_data.unique()[:sample_size]) # sample unique values

        summary.append({
            'Variable': col,
            'DataType': dtype,
            'UniqueValues': num_unique_no_sentinel,
            'Scenario': scenario,
            'SuggestedHandling': handling,
            'SampleValues': sample_values
        })

    return pd.DataFrame(summary)
```

Appendix D

Step 6: Cleaning function for multi-class and mixed variables

```
def categorical_transform_S6(df): # Step6; Scenario -> Multi class categorical
    df_cleaned = df.copy()
    sent_set = set([str(x).lower() for x in string_sentinels]) | set([str(x).lower() for x in standardised_sentinels])
    numeric_sentinel_set = set(float(x) for x in numeric_sentinels)

    # Helper normaliser
    def _norm(tok):
        if tok is None:
            return None
        s = str(tok).strip().lower()
        if s in blanks:
            return ''
        return s

    # Words -> number map
    word_num_map = {
        'zero':0.0,'one':1.0,'two':2.0,'three':3.0,'four':4.0,'five':5.0,'six':6.0,'seven':7.0,'eight':8.0,'nine':9.0,
        'ten':10.0,'eleven':11.0,'twelve':12.0,'thirteen':13.0,'fourteen':14.0,'fifteen':15.0,'twenty':20.0,
```

```
    'thirty':30.0,'forty':40.0,'fifty':50.0,'sixty':60.0,'seventy':70.0,'eighty':80.0,'ninety':90.0,
    'a few':2.0,'a couple':2.0,'few':2.0,'once':1.0,'twice':2.0,'several':3.0,'couple':2.0
}
def words_to_num(token):
    if token is None:
        return None
    token = token.strip().lower()
    if token in word_num_map:
        return word_num_map[token]
    try:
        return float(token)
    except Exception:
        return None

# Regexes for parsing
RE_TIME = re.compile(r'^\s*(\d{1,2}):(\d{2})\s*$')
RE_PERCENT = re.compile(r'^\s*(-?\d+(?:\.\d+)?)\s*%\s*$')
RE_DIRECT_NUM = re.compile(r'^\s*-?\d+(?:\.\d+)?\s*$')
RE_RANGE = re.compile(r'^\s*(-?\d+(?:\.\d+)?)\s*[---/]\s*(-?\d+(?:\.\d+)?)\s*$')
RE_RANGE_TO = re.compile(r'^\s*(-?\d+(?:\.\d+)?)\s*(?:to|TO)\s*(-?\d+(?:\.\d+)?)\s*$')
RE_PLUS = re.compile(r'^\s*(-?\d+(?:\.\d+)?)\s*\+\s*$')
RE_GT = re.compile(r'^\s*(?:>=|>)\s*(-?\d+(?:\.\d+)?)\s*$')
RE_LT_ANCHOR = re.compile(r'^(?:<|<=)\s*-?\d+(?:\.\d+)?')
RE_NUM_FIND = re.compile(r'-?\d+(?:\.\d+)?')
RE_WORD_TOK = re.compile(r"[a-z']+(?:\s[a-z']+)?")
# RE_SURVEY_TIME = re.compile(r'^\s*(?:[a-z]+\s+)?(\d+(?:\.\d+)?)\s*(?:month|week|year)s?.*$', re.IGNORECASE)

def norm_return(v):
    try:
        vf = float(v)
    except Exception:
        return None
    if vf in numeric_sentinel_set:
        return 'SENTINEL'
    return vf

```

```python
    def try_parse_token(x):
        """Return float, 'SENTINEL', or None for unknown token x"""
        if x is None:
            return None

        # preserve numeric pre transformed values
        if isinstance(x, (int, float)):
            if float(x) in numeric_sentinel_set:
                return 'SENTINEL'
            return float(x)

        if isinstance(x, str):
            sx = str(x).strip()
            if (sx.startswith('NOT ') or sx.lower().startswith('not ')) \
                and 'applicable' not in sx.lower() \
                and not sx.lower().startswith('not none'):
                return 0.0
        s = _norm(x)

        if s in ('none', 'none of these', 'none of the these'): # typo variant too
            return 0.0
        if s.startswith('not none'):
            return 1.0
        # Extending sentinel detection for categorical parsing scenario
        if s == '':
            return 'SENTINEL'
        if 'not applicable' in s or 'does not have' in s or "don't receive" in s or "don\x92t receive" in s or 'unknown' in s:
            return 'SENTINEL'
        if s in sent_set or re.fullmatch(r'[^0-9a-z]+', s):
            return 'SENTINEL'
        if s in ('other', 'other specify') or s.startswith('other (') or s.endswith('(specify)'):
            return 'SENTINEL'

```

```python
        s0 = s.replace(',', '').replace('$', '').replace(EUR, '')
           .replace('#', '')

        s0 = re.sub(r'(\d+)-', r'\1 -', s0)
        s0 = re.sub(r'-(\d+)', r'- \1', s0)

        RE_PREFIX_NUM = re.compile(r'^\s*(-?\d+(?:\.\d+)?)\.\s*.+
           $')

        m = RE_PREFIX_NUM.match(s0)
        if m:
            # If the format is 'NUM. anything', return the number
                as the category value
            return norm_return(float(m.group(1)))

        # time
        m = RE_TIME.match(s0)
        if m:
            h, mi = float(m.group(1)), float(m.group(2))
            return norm_return(h * 60.0 + mi)

        # percent
        m = RE_PERCENT.match(s0)
        if m:
            return norm_return(float(m.group(1)))

        # direct numeric
        if RE_DIRECT_NUM.match(s0):
            return norm_return(s0)

        # ranges
        m = RE_RANGE.match(s0)
        if m:
            try:
                a, b = float(m.group(1)), float(m.group(2))
                return norm_return((a + b) / 2.0)
            except Exception:
                pass
        m = RE_RANGE_TO.match(s0)
        if m:
            try:
```

```
                a, b = float(m.group(1)), float(m.group(2))
                return norm_return((a + b) / 2.0)
            except Exception:
                pass

        # plus / greater than
        m = RE_PLUS.match(s0) or RE_GT.match(s0)
        if m:
            nums = RE_NUM_FIND.findall(s0)
            if nums:
                return norm_return(nums[0])

        # less than
        if RE_LT_ANCHOR.search(s0) or 'less than' in s0:
            nums = RE_NUM_FIND.findall(s0)
            if nums:
                return norm_return(nums[0])

        # or less / more
        m = re.search(r'(-?\d+(?:\.\d+)?)\s*or\s*less', s0)
        if m:
            return norm_return(m.group(1))
        m = re.search(r'(-?\d+(?:\.\d+)?)\s*or\s*more', s0)
        if m:
            return norm_return(m.group(1))

        # "more than X but less than Y", "over X but less than Y"
        m = re.search(r'(?:more than|over)\s*e?(-?\d+(?:\.\d+)?)\s*(?:but\s*)?(?:less than|under)\s*e?(-?\d+(?:\.\d+)?)', s0)
        if m:
            try:
                a, b = float(m.group(1)), float(m.group(2))
                return norm_return((a + b) / 2.0)
            except Exception:
                pass

        # "between X and Y" / "between X - Y"
        m = re.search(r'between\s*e?(-?\d+(?:\.\d+)?)\s*(?:and|[---/])\s*e?(-?\d+(?:\.\d+)?)', s0)
        if m:
```

```
            try:
                a, b = float(m.group(1)), float(m.group(2))
                return norm_return((a + b) / 2.0)
            except Exception:
                pass

        # survey time based
        time_map_weeks = {
            'one week': 1.0,
            'two weeks': 2.0,
            'a month (4 weeks)': 4.0,
            'a month/4 weeks': 4.0,
            'three months (13 weeks)': 13.0,
            'three months/13 weeks': 13.0,
            'six months (26 weeks)': 26.0,
            'six months/26 weeks': 26.0,
            'one year (12 months/52 weeks)': 52.0,
            'one year/12 months/52 weeks': 52.0
        }
        if s0 in time_map_weeks:
            return norm_return(time_map_weeks[s0])

        # If a category is '3 months or more', try to extract the number and use upper bound
        if 'months or more' in s0 and RE_NUM_FIND.search(s0):
            nums = RE_NUM_FIND.findall(s0)
            if nums:
                months = float(nums[0])
                return norm_return(months * 4.33) # Convert months to weeks (4.33 weeks per month)

        # Frequency calendar based
        if any(tok in s0 for tok in ('day','week','month','year','daily','never','almost every day','once','twice','thrice','three','several','couple')):
            nums = RE_NUM_FIND.findall(s0)
            if nums:
                val = (float(nums[0]) + float(nums[1])) / 2.0 if len(nums) >= 2 else float(nums[0])
            else:
                if 'once or twice' in s0:
```

```
                    val = 1.5
                else:
                    val = None
                    for w in RE_WORD_TOK.findall(s0):
                        wn = words_to_num(w)
                        if wn is not None:
                            val = wn
                            break
            if val is not None:
                if 'day' in s0:
                    if 'week' in s0:
                        return norm_return(val * 4.33)
                    if 'month' in s0:
                        return norm_return(val)
                    if 'daily' in s0 or 'every day' in s0 or '
                       almost every day' in s0:
                        return norm_return(val * 30.0)
                    return norm_return(val * 4.33)
                if 'week' in s0:
                    return norm_return(val * 4.33)
                if 'month' in s0:
                    return norm_return(val)
                if 'year' in s0:
                    return norm_return(val / 12.0)
                return norm_return(val)


        likert_map = {
            # Likert / ordinals (expanded with quintile/lowest/
               highest detection)
            'strongly disagree': 0.0, 'disagree': 1.0, 'neither
               agree nor disagree': 2.0, 'neither': 2.0,
            'agree': 3.0, 'strongly agree': 4.0,
            'very poor': 0.0, 'poor': 1.0, 'fair': 2.0, 'good':
               3.0, 'very good': 4.0, 'excellent': 5.0,
            'not at all': 0.0, 'hardly ever': 0.0,
            'a little': 1.0, 'some': 2.0, 'a lot': 3.0, 'alot':
               3.0, 'often': 3.0,
            'yes': 1.0, 'no': 0.0, 'sometimes': 2.0, 'rarely':
               1.0,
            'pass': 1.0, 'fail': 0.0, '100': 100.0,
```

```python
        # Problem severity mappings
        'no problem': 0.0, 'minor problem': 1.0, 'moderate problem': 2.0, 'major problem': 3.0,
        # Concern level mappings
        'not at all concerned': 0.0,'somewhat concerned': 1.0,'fairly concerned': 2.0,'very concerned': 3.0,
        # Survey frequency mappings
        'daily / almost daily': 7.0, 'once a week or more': 6.0,'twice a month or more': 5.0,'about once a month': 4.0,
        'every few months': 3.0, 'about once or twice a year': 2.0,'less than once a year': 1.0,'never': 0.0,
        'daily': 7.0,'every day': 7.0,'almost every day': 7.0,'most or all of the time': 7.0,
        'every week': 6.0,'once a week': 6.0,'several times per week': 6.0,
        'twice a month': 5.0,'two a month': 5.0,'several times per month': 5.0,'once or twice a month': 5.0,
        'once a month': 4.0, 'once or twice a year': 2.0,'less than once a month': 1.0,
    }

    if s0 in likert_map:
        return float(likert_map[s0])

    _CONJ_RE = re.compile(r'\b(?:and|but|or|because|however|although|answered)\b')
    _NO_SPECIFIC = ['problem', 'response', 'prompt']

    if s0.startswith('no ') and not any(w in s0 for w in _NO_SPECIFIC) and not _CONJ_RE.search(s0):
        return 0.0

    if s0.startswith('yes ') and not _CONJ_RE.search(s0):
        return 1.0

    # Quintile / ordinal like tokens: 'lowest', '2nd quintile', 'highest'
    if 'quintile' in s0 or 'lowest' in s0 or 'highest' in s0 or 'middle' in s0 or 'median' in s0:
```

```
                if 'lowest' in s0:
                    return norm_return(1.0)
                if 'highest' in s0:
                    return norm_return(5.0)
                if 'middle' in s0 or 'median' in s0:
                    return norm_return(3.0)
                mq = re.search(r'(\d+)(?:st|nd|rd|th)?\s+quintile', s0)
                if mq:
                    return norm_return(float(mq.group(1)))
                nums = RE_NUM_FIND.findall(s0)
                if nums:
                    return norm_return(nums[0])

        if 'adjacent' in s0: # Adjacent boxes: keep semantic average if present
            nums = RE_NUM_FIND.findall(s0)
            if len(nums) >= 2:
                return norm_return((float(nums[0]) + float(nums[1])) / 2.0)
            elif nums:
                return norm_return(nums[0])
            return None

        # Fallback: first number
        nums = RE_NUM_FIND.findall(s0)
        if nums:
            return norm_return(nums[0])

        return None

    def category_sort_key(u):
        us = _norm(u) or ''
        parsed = try_parse_token(us)
        if isinstance(parsed, (int, float)):
            return (0, float(parsed), us)
        if us == 'no' or us.startswith('no ') or us.startswith('not '):
            return (1, 0.0, us)
        if 'yes' in us:
            if 'both' in us or 'two' in us:
```

```
            return (1, 2.0, us)
        if 'one' in us:
            return (1, 1.0, us)
        return (1, 1.0, us)
    amount_keywords = ['none', 'not at all', 'hardly ever', 'a little', 'some', 'a lot', 'alot', 'often', 'sometimes', 'rarely']
    for i, key in enumerate(amount_keywords):
        if key in us:
            return (2, float(i), us)
    if 'adjacent boxes' in us or 'adjacent numbers' in us:
        nums = RE_NUM_FIND.findall(us)
        if len(nums) >= 2:
            a, b = float(nums[0]), float(nums[1])
            return (10, (a + b) / 2.0, us)
        elif nums:
            return (10, float(nums[0]), us)
        return (10, 0, us)
    return (99, 0, us)

# Operate only on categorical columns
cat_cols = df_cleaned.select_dtypes(include='category').columns.tolist()
for col in cat_cols:
    uniques = list(dict.fromkeys(df_cleaned[col].tolist())) # Preserve original tokens with types (do NOT cast to str here)
    normalised_pairs = [(u, _norm(u)) for u in uniques] # Build normalised pairs keeping original token too
    # Keep tokens that are non sentinel and parseable (strings or numbers)
    non_sentinel = [u for u, nu in normalised_pairs if (nu and nu not in sent_set and nu not in blanks and try_parse_token(nu) != 'SENTINEL')]

    if not non_sentinel:
        continue

    # Computing parse results keyed by normalized string (but keep list of originals too)
```

```python
        parse_results = { _norm(u): try_parse_token(_norm(u)) for u in non_sentinel }

        numeric_count = sum(1 for v in parse_results.values() if isinstance(v, (int, float)))
        sentinel_parsed_count = sum(1 for v in parse_results.values() if v == 'SENTINEL')
        all_parse_numeric = (numeric_count + sentinel_parsed_count == len(parse_results))

        if all_parse_numeric:
            # Map token string -> numeric (SENTINEL -> -1.0)
            code_map = { k: (float(v) if isinstance(v, (int, float)) else -1.0) for k, v in parse_results.items() }

            def map_to_numeric(x):
                # Preserve the already numeric cells (except numeric sentinels)
                if isinstance(x, (int, float)) and not pd.isna(x):
                    if float(x) in numeric_sentinel_set:
                        return -1.0
                    return float(x)
                if pd.isna(x):
                    return -1.0
                xs = _norm(x)
                if not xs or xs in sent_set or xs in blanks or re.fullmatch(r'[^0-9a-z]+', xs):
                    return -1.0
                return float(code_map.get(xs, -1.0))

            df_cleaned[col] = df_cleaned[col].apply(map_to_numeric).astype(float)

        else:
            first_seen_norm = []
            seen = set()
            for u in non_sentinel:
                nu = _norm(u)
                if nu not in seen:
```

```
                first_seen_norm.append(nu)
                seen.add(nu)

        parse_results = {nu: try_parse_token(nu) for nu in
            first_seen_norm}
        max_reserved_val = -1.0
        for nu, val in parse_results.items():
            if isinstance(val, (int, float)):
                max_reserved_val = max(max_reserved_val, val)
                    # Assuming any parsed float is a reserved
                    ID if it came from the map
        start_i = int(max_reserved_val) + 1
        if start_i == 0: # Ensure we never start at 0 if 0 is
            a reserved category i.e for ('no')
            start_i = 1
        string_tokens = [nu for nu in first_seen_norm if not
            isinstance(parse_results.get(nu), (int, float))]
        sorted_strings = sorted(string_tokens, key=
            category_sort_key)

        code_map_strings = { norm_tok: i for i, norm_tok in
            enumerate(sorted_strings, start=start_i) }

        def map_mixed(x):
            if isinstance(x, (int, float)) and not pd.isna(x)
                :# Preserve real numeric cell values
                if float(x) in numeric_sentinel_set:
                    return -1.0
                return float(x)
            if pd.isna(x):
                return -1.0
            xs = _norm(x)
            if xs == '' or xs in sent_set or xs in blanks or
                re.fullmatch(r'[^0-9a-z]+', xs):
                return -1.0

            parsed_val = try_parse_token(xs)
            if parsed_val == 'SENTINEL':
                return -1.0
            if isinstance(parsed_val, (int, float)):#
                Captures 'no' (0.0), 'yes' (1.0), and other
```

```
                    hardcoded likert values.
                     return float(parsed_val)
                if xs in code_map_strings:
                     return float(code_map_strings[xs])
                return -1.0

            df_cleaned[col] = df_cleaned[col].apply(map_mixed).
                astype(float)

    # Final sweep on numeric columns
    for c in df_cleaned.select_dtypes(include='float').columns:
        df_cleaned[c] = df_cleaned[c].fillna(-1.0)
        df_cleaned[c] = df_cleaned[c].replace(list(
            numeric_sentinel_set), -1.0)

    return df_cleaned
```

Appendix E

Main cleaning pipeline execution script

```
# Cleaning pipeline to all waves
dfs_cleaned = {}

# flattened dtype df
dfs_flat = {wave: flatten_column_types(df) for wave, df in dfs.
    items()} # flatten dtypes

# Original df
display(scenario_count_table(dfs_flat).style.set_caption("--- 
    Original Table - pre cleaning ---")) # display original
    scenario count table

# Step 1: Clean empty/missing values
dfs_step1 = {wave: clean_empty_missing(df) for wave, df in
    dfs_flat.items()} # apply step 1 cleaning
display(scenario_count_table(dfs_step1).style.hide(axis="index").
    set_caption("--- Step 1: Empty / Blank - post cleaning ---"))

# Step 2: Clean sentinel values
dfs_step2 = {wave: clean_sentinel(df) for wave, df in dfs_step1.
    items()} # apply step 2 cleaning
display(scenario_count_table(dfs_step2).style.hide(axis="index").
    set_caption("--- Step 2: Sentinel - post cleaning ---"))

```

```python
# Step 3: Numerical Transformation
dfs_step3 = {wave: numerical_transform(df) for wave, df in dfs_step2.items()} # apply step 3 cleaning
display(scenario_count_table(dfs_step3).style.hide(axis="index").set_caption("--- Step 3: Numeric Transformation - post cleaning ---"))

# Step 4: Categorical Transformation for censored scenario
dfs_step4 = {wave: categorical_transform_S4(df) for wave, df in dfs_step3.items()} # apply step 4 cleaning
display(scenario_count_table(dfs_step4).style.hide(axis="index").set_caption("--- Step 4: Category Transformation S4 - post cleaning ---"))

# Step 5: Categorical Transformation for binary values
dfs_step5 = {wave: binary_transform_S5(df) for wave, df in dfs_step4.items()} # apply step 5 cleaning
display(scenario_count_table(dfs_step5).style.hide(axis="index").set_caption("--- Step 5: Binary Transformation S5 - post cleaning ---"))

# Step 6: Categorical Transformation for multi class/mixed values
dfs_step6 = {wave: categorical_transform_S6(df) for wave, df in dfs_step5.items()} # apply step 6 cleaning
display(scenario_count_table(dfs_step6).style.hide(axis="index").set_caption("--- Step 6: Category Transformation S6 - post cleaning ---"))

dfs_cleaned = dfs_step6
```

Appendix F

Grouped variable audit script (Markdown output)

```
def show_grouped_vars_for_scenarios(dfs_original, dfs_transformed, scenarios_to_show): # Function to show grouped variables per scenario across waves -> markdown format
    """
␣␣␣␣Used␣to␣inspect␣the␣scenario␣variables/features␣that␣need␣handling␣for␣transformation/cleaning.
␣␣␣␣A␣pre␣and␣post␣cleaning␣helper␣function␣to␣identify␣variables␣that␣need␣specific␣handling␣and␣to
␣␣␣␣Inspect␣actual␣changes␣for␣a␣scenario␣from␣previous␣cleaning␣step␣to␣current␣step.
␣␣␣␣"""

    print("#␣Category␣Transformation␣Mapping")
    # print(f"\n## Scenarios: {scenarios_to_show}") # Markdown mapping dictionary header
    for wave in dfs_original.keys():
        print(f"\n###␣====================␣{wave.upper()}␣====================")

        mapping_df = detect_column_scenario_mapping(dfs_transformed[wave]) # Get scenario mapping from transformed/cleaned to see overall changes for all variables
```

```python
        # mapping_df = detect_column_scenario_mapping(dfs_original[wave]) # Get scenario mapping from original to see untransformed values

        for scenario in scenarios_to_show:
            vars_in_scenario = mapping_df[mapping_df['Scenario'] == scenario]['Variable'].tolist() # Variables in scenario
            print(f"\n--- Scenario {scenario}: {len(vars_in_scenario)} ---") # Scenario header for pipeline git commit
            if not vars_in_scenario:
                print("\nNo variables found.")
                continue

            # Dictionary: frozen set(values) -> list of variables
            groups = {}

            for var in vars_in_scenario:
                unique_vals = list(dict.fromkeys(dfs_original[wave][var].dropna()))  # Preserve first seen order from ORIGINAL
                key = tuple(unique_vals)  # Use ordered tuple as key

                if key not in groups: # Initialise list if key not present
                    groups[key] = []
                groups[key].append(var) # Append variable to the group

            # Print grouped variables in markdown format friendly for Notion -> mapping dictionary
            for value_set, var_list in groups.items():
                print("\n*Variables:*","`" + ", ".join(var_list) + "`  ") # \n
                print(f"*Pre transformed values:* {list(value_set)}  ")

                # Pull transformed values from dfs_cleaned / specific step using preserved order
```

```
                try:
                    df_before = dfs_original[wave][var_list[0]]
                    df_after = dfs_transformed[wave][var_list[0]]

                    # Build transformed list in same order as original values
                    transformed_vals = []
                    for orig_val in value_set: # Iterate through original values in order
                        mask = df_before == orig_val
                        if mask.any():
                            trans_vals = df_after[mask].unique().tolist() # Get unique transformed values
                            if trans_vals:
                                transformed_vals.append(trans_vals[0]) # Append first transformed value
                except Exception:
                    transformed_vals = []
                print(f"*Transformed numeric:* `{transformed_vals}`")


scenarios_to_show = SCENARIOS
# scenarios_to_show = ["Clean numeric"]
# scenarios_to_show = ["Censored / bracketed numeric"]
# scenarios_to_show = ["Binary categorical"]
# scenarios_to_show = ["Multi class categorical / mixed"]

show_grouped_vars_for_scenarios(dfs, dfs_cleaned, scenarios_to_show) # prev step == comparison to current step else current step transformed/cleaned for ready .md dictionary
```

Appendix G

Sample Markdown audit output for variable transformations

The grouped Markdown representation reveals three critical points:

1. Semantically identical categories are correctly grouped even under different variable names (`ph001` vs `ph009`), enabling direct comparison of representations that would otherwise appear fragmented across waves.

2. Numerical encoding is invariant to category ordering, preserving ordinal meaning across waves. Without this approach, positional encoding would assign inconsistent values, potentially distorting performance for downstream models.

3. For variables with a large and heterogeneous set of observed values (e.g., income bands, frequency ranges, or mixed numeric text responses), the grouped view provided essential visibility into how complex category spaces were being collapsed into stable numeric representations, supporting both validation and error detection at scale.

```
%----------------------------------------------------------------
↪  ------%
%------------------------MARKDOWN_MAPPING_DICTIONARY----------------
↪  ------%
%----------------------------------------------------------------
↪  ------%

# Category Transformation Mapping

### =================== WAVE1 ====================

--- Scenario Empty / blank values: 0 ---

No variables found.

--- Scenario Numeric with sentinel: 0 ---

No variables found.

--- Scenario Clean numeric: 809 ---

*Variables:* 'sex, gd002'
*Pre transformed values:* ['Male', 'Female']
*Transformed numeric:* '[1.0, 0.0]'

*Variables:* 'ph001'
*Pre transformed values:* ['Fair', 'Good', 'Very good', 'Excellent',
↪  'Poor', "Don't know"]
*Transformed numeric:* '[2.0, 3.0, 4.0, 5.0, 1.0, -1.0]'

*Variables:* 'bh006'
*Pre transformed values:* ['20-39', 10.0, -1.0, 15.0, 5.0, '60+',
↪  "Don't Know", 2.0, 8.0, 3.0, '40-59', 4.0, 7.0, 1.0, 6.0, 14.0,
↪  12.0, 16.0, 18.0, 13.0, 9.0, 0.0, 19.0, 11.0, 17.0]
```

```
*Transformed numeric:* '[29.5, 10.0, -1.0, 15.0, 5.0, 60.0, -1.0,
↪  2.0, 8.0, 3.0, 49.5, 4.0, 7.0, 1.0, 6.0, 14.0, 12.0, 16.0, 18.0,
↪  13.0, 9.0, 0.0, 19.0, 11.0, 17.0]'

--- Scenario Censored / bracketed numeric: 0 ---

No variables found.

--- Scenario Binary categorical: 0 ---

No variables found.

--- Scenario Multi class categorical / mixed: 0 ---

No variables found.

--- Scenario Single value categorical: 0 ---

No variables found.

--- Scenario Special symbol / mixed: 0 ---

No variables found.

--- Scenario Unknown / complex mixed: 0 ---

No variables found.

### ==================== WAVE2 ====================

...

--- Scenario Clean numeric: 482 ---

*Variables:* 'gd002, sex'
*Pre transformed values:* ['Female', 'Male']
```

```
*Transformed numeric:* '[0.0, 1.0]'

*Variables:* 'bh006'
*Pre transformed values:* [-1.0, '10 - 19', '20 - 29', 'Less than
↪ 10', '30+', "Don't know"]
*Transformed numeric:* '[-1.0, 14.5, 24.5, 10.0, 30.0, -1.0]'

*Variables:* 'ph001'
*Pre transformed values:* ['Very good', 'Good', 'Fair', 'Excellent',
↪ 'Poor']
*Transformed numeric:* '[4.0, 3.0, 2.0, 5.0, 1.0]'

...

### ==================== WAVE3 ====================

...

--- Scenario Clean numeric: 639 ---

*Variables:* 'sex, gd002'
*Pre transformed values:* ['Female', 'Male']
*Transformed numeric:* '[0.0, 1.0]'

*Variables:* 'ph001, ph009'
*Pre transformed values:* ['Very Good', 'Good', 'Excellent', 'Fair',
↪ 'Poor', 'DK']
*Transformed numeric:* '[4.0, 3.0, 5.0, 2.0, 1.0, -1.0]'

*Variables:* 'bh006'
*Pre transformed values:* [-1.0, '10 - 19', 'Less than 10', '20 -
↪ 29', '30+', -98.0, -99.0]
*Transformed numeric:* '[-1.0, 14.5, 10.0, 24.5, 30.0, -1.0, -1.0]'

...
```

G. SAMPLE MARKDOWN AUDIT OUTPUT FOR VARIABLE TRANSFORMATIONS

```
### ==================== WAVE4 ====================

...

--- Scenario Clean numeric: 592 ---

*Variables:* 'bh006'
*Pre transformed values:* [-1.0, 'Less than 10', '20 - 29', '10 -
↪  19', '30+', "Don't know"]
*Transformed numeric:* '[-1.0, 10.0, 24.5, 14.5, 30.0, -1.0]'

*Variables:* 'gd002, sex'
*Pre transformed values:* ['Female', 'Male']
*Transformed numeric:* '[0.0, 1.0]'

*Variables:* 'ph001'
*Pre transformed values:* ['Good', 'Fair', 'Poor', 'Excellent', 'Very
↪  Good', 'RF', 'DK']
*Transformed numeric:* '[3.0, 2.0, 1.0, 5.0, 4.0, -1.0, -1.0]'

...

### ==================== WAVE5 ====================

...

--- Scenario Clean numeric: 591 ---

*Variables:* 'bh006'
*Pre transformed values:* [-1.0, 'Less than 10', '20 - 29', '10 -
↪  19', '30+', "Don't know"]
*Transformed numeric:* '[-1.0, 10.0, 24.5, 14.5, 30.0, -1.0]'

*Variables:* 'gd002, sex'
*Pre transformed values:* ['Female', 'Male']
*Transformed numeric:* '[0.0, 1.0]'
```

```
*Variables:* 'ph001'
*Pre transformed values:* ['Very Good', 'Fair', 'Excellent', 'Good',
↪  'Poor']
*Transformed numeric:* '[4.0, 2.0, 5.0, 3.0, 1.0]'

...
```

Appendix H

Track B cohort sample sizes and incident rates

Table H.1: Cohort sizes and incident rates for the 24 disease specific outcomes in Track B.

Disease	**Type**	**Obs.** n	**Inc.** n	**Inc. rate %**
Circulatory disease	Clinical	14539	1774	12.2
Endocrine/metabolic diseases	Clinical	4117	362	8.8
Eye diseases	Clinical	4396	444	10.1
Musculoskeletal diseases	Clinical	4269	277	6.5
Functional decline	Functional	14296	3217	22.5
Fear of falling	Functional	18114	2536	14.0
Diabetes	Family history	4307	276	6.4
High cholesterol	Family history	4312	517	12.0
Hypertension	Family history	3590	679	18.9
Heart disease	Family history	2770	598	21.6
Osteoporosis	Family history	4779	330	6.9
Alzheimer's/dementia	Family history	4827	309	6.4
Cancer	Family history	3971	532	13.4
Depression	Family history	4760	343	7.2
Pain analgesia	Medication	2152	775	36.0
High cholesterol	Medication	733	181	24.7
Antidepressant	Medication	4378	140	3.2
Antihypertensive	Medication	19845	893	4.5
Chronic pain (proxy)	Self-reported disease	15571	2709	17.4
Hearing loss	Self-reported disease	10780	1628	15.1
Urinary incontinence	Self-reported disease	16208	1361	8.4
Chronic pain	Self-reported disease	10914	1888	17.3
Angioplasty/stent	Procedure	9233	129	1.4
Hypertension	Doctor diagnosis	250	104	41.6

Appendix I

Item index and scoring of individual features

Table I.1: Scoring scheme for the 40 individual feature items.

Item	Variable	Scoring
Doctor diagnosed eye disease	ph105_5	0 = None; 1 = Yes
Doctor diagnosed hypertension	ph201_01	1 = Yes; 0 = No
Doctor diagnosed high cholesterol	ph201_08	0 = No; 1 = Yes
No conditions	ph201_14	1 = None; 0 = Otherwise
Doctor diagnosed arthritis	ph301_03	1 = Yes; 0 = No
No listed conditions	ph301_17	1 = None; 0 = Otherwise
Self-reported long-term chronic/health problem	ph003	1 = Yes; 0 = No; -1 = DK/RF
Long term illness/health problem	DISlongterm	0 = No illness; 1 = Illness; 1 = Limiting illness
Neoplasms	ICD10_02	0 = No; 1 = Yes
Diseases of the blood	ICD10_03	0 = No; 1 = Yes
Endocrine/metabolic disease	ICD10_04	0 = No; 1 = Yes
Mental and behavioural disorders	ICD10_05	0 = No; 1 = Yes
Diseases of the nervous system	ICD10_06	0 = No; 1 = Yes
Eye disease	ICD10_07	0 = No; 1 = Yes

Continued on next page

Item	Variable	Scoring
Circulatory disease	ICD10_09	0 = No; 1 = Yes
Respiratory disease	ICD10_10	0 = No; 1 = Yes
Digestive system disease	ICD10_11	0 = No; 1 = Yes
Musculoskeletal disease	ICD10_13	0 = No; 1 = Yes
Genitourinary disease	ICD10_14	0 = No; 1 = Yes
Family history of diabetes	ph726_01	0 = No; 1 = Yes
Family history of high cholesterol	ph726_02	0 = No; 1 = Yes
Family history of hypertension	ph726_03	0 = No; 1 = Yes
Family history of heart disease	ph726_04	0 = No; 1 = Yes
Family history of osteoporosis	ph726_06	0 = No; 1 = Yes
Family history of alzheimer's / dementia	ph726_07	0 = No; 1 = Yes
Family history of cancer	ph726_08_to_11	1 = Yes; 0 = No
Family history of depression	ph726_12	0 = No; 1 = Yes
Family history of other condition	ph726_95	0 = No; 1 = Yes
No family history of any disease	ph726_96	0 = No; 1 = Yes
Pain medication use	ph505	1 = Yes; 0 = No; -1 = DK/RF
Cholesterol medication	ph225	1 = Yes; 0 = No; -1 = DK/RF
Antidepressant use	MDantidepressant	0 = No; 1 = Yes
Antihypertensive use	MDantihypertensives	0 = No; 1 = Yes
Self-rated chronic pain	ph501 (proxy)	1 = Yes; 0 = No; -1 = DK
Self-rated chronic pain	CHRany_pain	0 = No; 1 = Yes
Self-rated hearing loss	ph145	0 = No; 1 = Yes; -1 = DK
Self-rated urinary incontinence	ph601	0 = No; 1 = Yes; -1 = DK/RF
Angioplasty/stent	ph208	0 = No; 1 = Yes; -1 = DK
Fear of falling	ph408	0 = No; 1 = Yes; -1 = DK
Antihypertensive medication use	ph202a	1 = Yes; 0 = No; -1 = DK

Appendix J

Top 20 feature importance for Track A designs D1 and D2 models

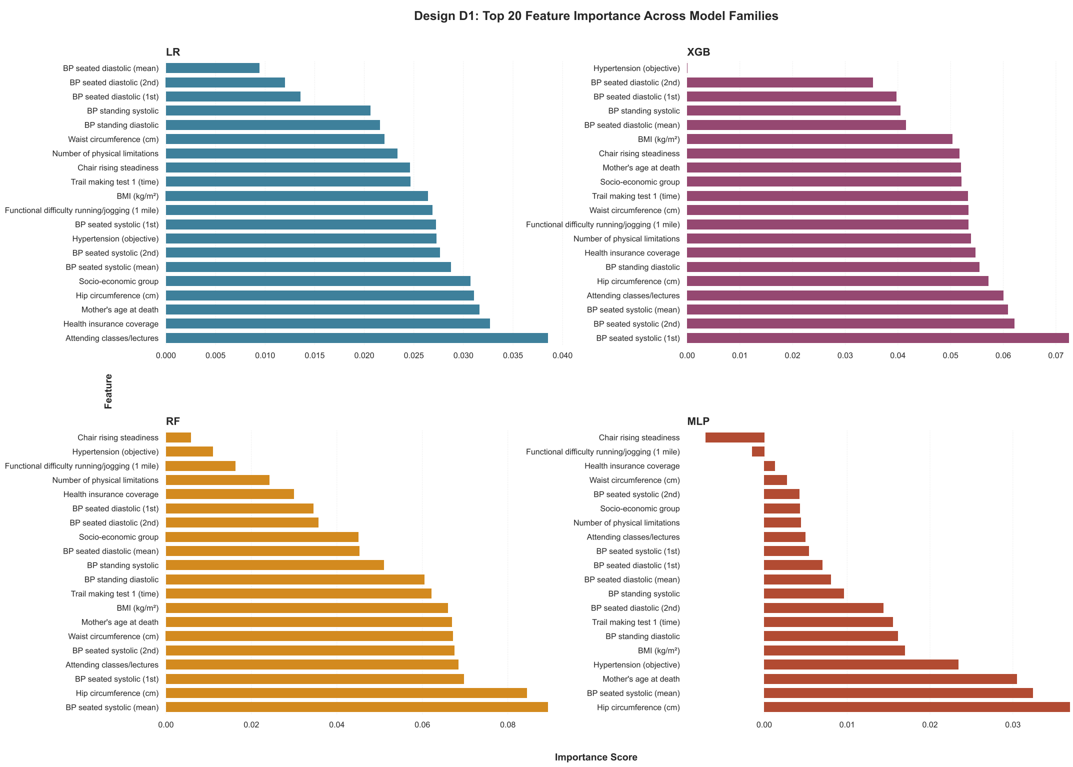

Figure J.1: Top 20 features by importance for D1.

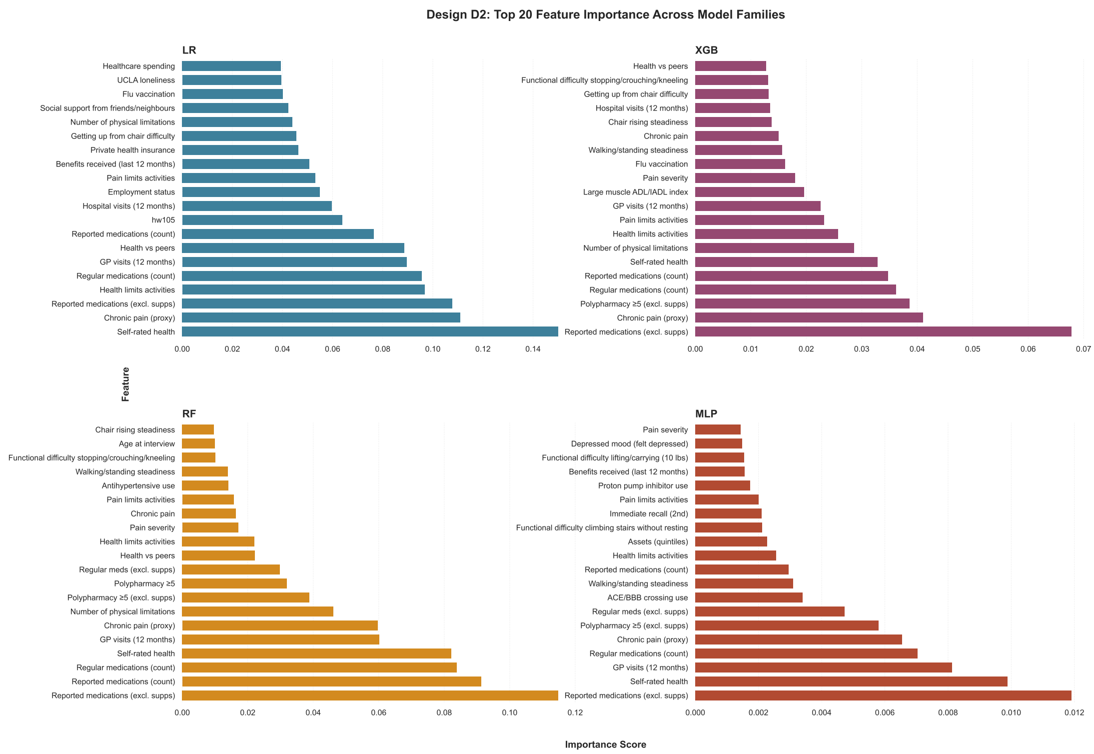

Figure J.2: Top 20 features by importance for D2.

Appendix K

SHAP value plots for Track A and Track B

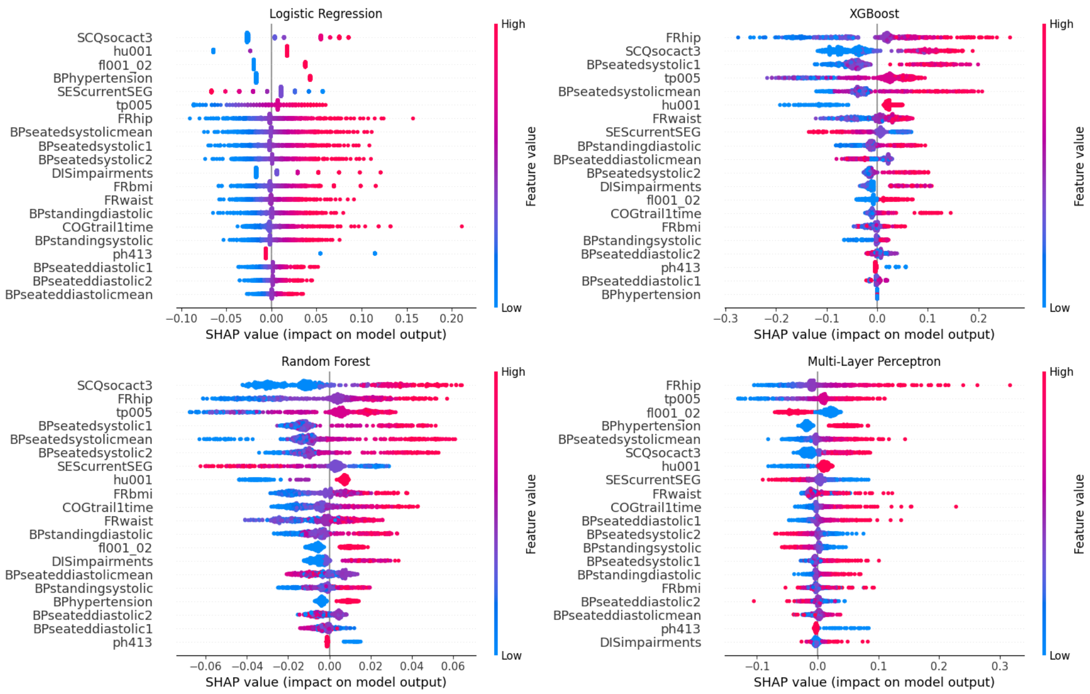

Figure K.1: Global SHAP beeswarm plots for D1 across all four model families (top-20 features by mean |SHAP|).

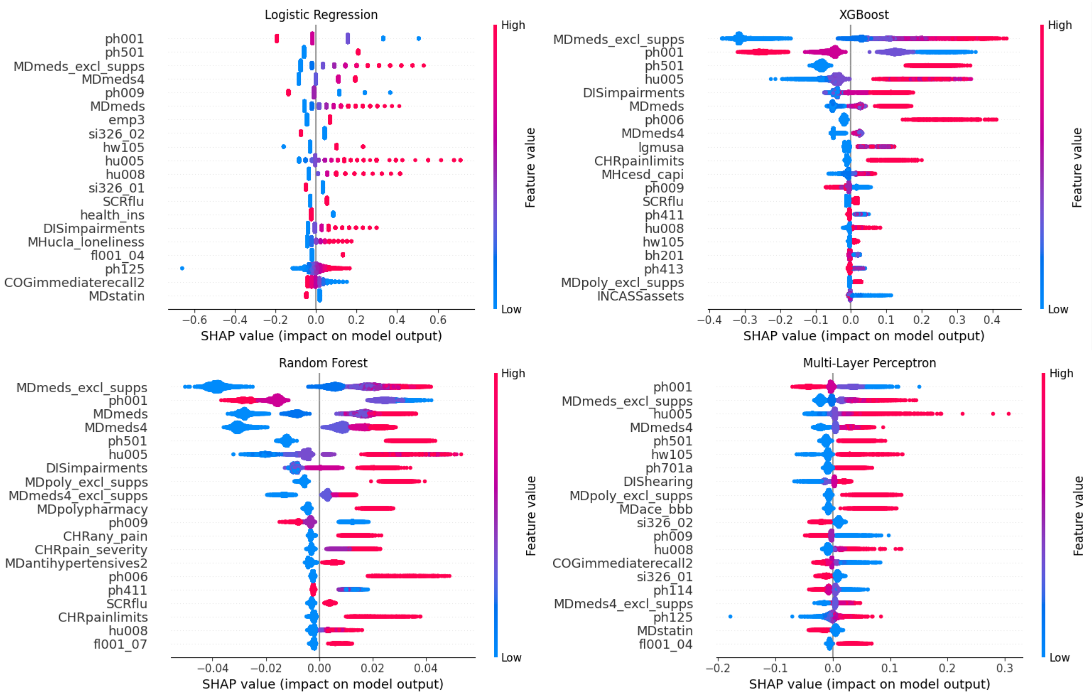

Figure K.2: Global SHAP beeswarm plots for D2 across all four model families (top-20 features by mean |SHAP|).

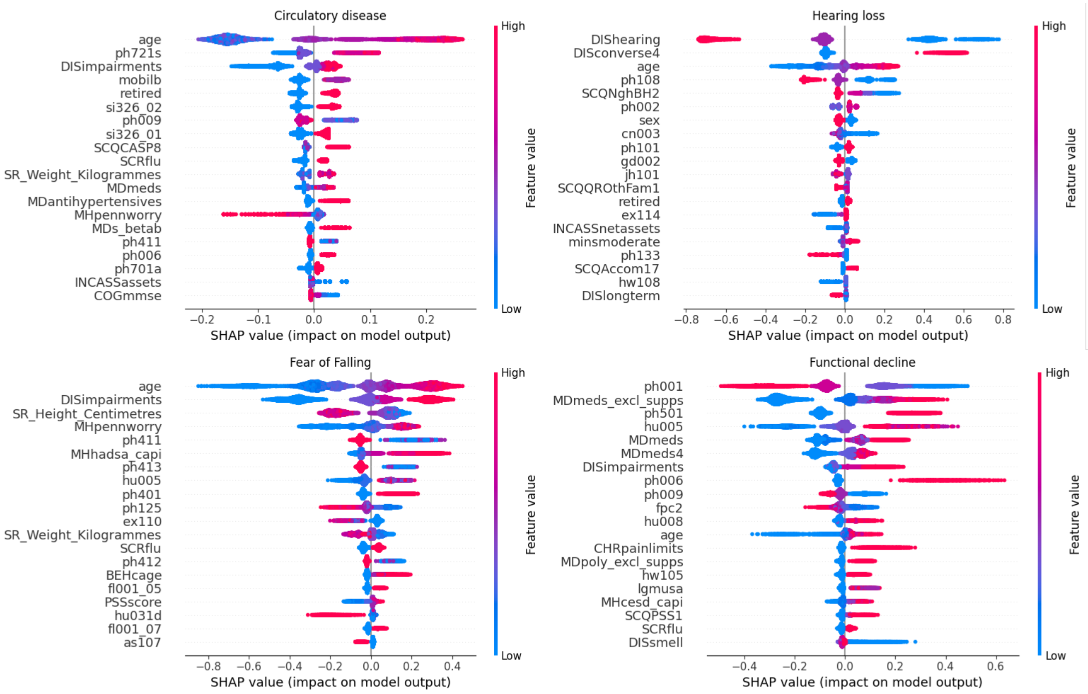

Figure K.3: Global SHAP beeswarm plots for four Track B disease specific XGB models: functional decline, fear of falling, hearing loss, and circulatory disease.

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